Green Innovation Park Renovation Project with Emphasis On Integrated Design Process

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Abstract

Keywords: Integrated Design Process (IDP), Air Handling Unit (AHU), Daylight, Ventilation, Heating.

This project focuses on the sustainable renovation of the Green Innovation Park building located in Alnarp, Southern Sweden, on the campus of the Swedish University of Agricultural Sciences. The primary goal is to enhance daylight performance, energy efficiency, overall indoor environment, and accessibility to the building, with a strong emphasis on the Integrated Design Process (IDP). Simulation tools, such as Climate Studio, Ladybug tools, Rhino, Revit, and Dialux, were used to analyse the building's systems in depth, as well as its daylight performance, energy consumption, thermal comfort, and ventilation performance.

The Existing building did not fulfill current daylight performance standards or energy performance standards, while having a claustrophobic feel inside, with prevalent outdated design strategies. The proposed renovation strategies include adding insulation to some existing components of the building, as well as a complete retrofit of all windows present on the building. Moreover, passive and active exterior shading strategies are implemented to reduce glare inside the building while preventing excessive thermal heat gains and subsequent overheating of interior spaces.

Building after renovation demonstrates improvements in all aspects of analysis. Spatial daylight autonomy exceeds 75%, and the mean daylight factor is above the required 1%. Moreover, EP_{pet} primary energy consumption of the building reaches roughly 48 kWh/m^2/year while overheating was kept under 1% in the April to September timespan.

1. Introduction

With increasing climate challenges and environmental pressures on the building sector, existing buildings have become a key focus for efforts to improve energy efficiency and reduce emissions. International estimates indicate that buildings consume more than a third of global energy and produce approximately 37% of total energy-related carbon emissions[1].

Multiple studies have shown that improving the use of daylighting in buildings can reduce lighting energy consumption by 20% to 60%, especially when incorporating smart control systems that adjust electric lighting based on daylight availability [2].

In addition to reducing energy consumption, daylighting design plays a fundamental role in enhancing indoor environmental quality in terms of visual comfort and light distribution. For this reason, the project follows the guidelines set forth in the standard SS-EN 17037:2018+A1:2021, which sets precise requirements for providing adequate daylighting in interior spaces. The standard stipulates that daylighting should cover at least 50% of the room area with an illumination level of at least 300 lux, and 95% with an illumination level of at least 100 lux, during half of the annual light hours. The standard also includes additional considerations such as the quality of window views, exposure to direct sunlight, and glare control, making daylighting a functional and aesthetic part of the design, rather than simply an energy factor [3].

On the other hand, the importance of good ventilation cannot be overlooked, as indoor air quality is directly linked to the health and comfort of occupants, especially in enclosed workplaces. Swedish regulators recommend an outdoor air supply of at least 0.35 l/s/ m^2 + 7 l/s/person, with additional requirements based on the space size.

1.1 *Project Description and Purpose*

This academic project is being implemented as a practical example for the renovation of an office building known as the "Green Innovation Park" located in Alnarp, Sweden while maintaining its primary function. The project aims to improve the building's environmental performance by rethinking ventilation systems, distributing daylight, and achieving high thermal comfort for occupants. The focus is not only on reducing consumption, but also on creating a healthier and more sustainable indoor environment.

The project also relies on integrating different disciplines from the early stages of design, known as an Integrated Design Process (IDP). This approach doesn't focus solely on partial solutions but rather seeks to create harmony between the various systems in the building, which impacts the overall energy performance and increases the reliability of the final result.

In addition to environmental standards, the project is designed to meet the Swedish Building Regulations (BBR), ensuring equitable use of the building by all groups, including people with disabilities. Therefore, building entrances must be barrier-free, with ramps or elevators provided where necessary. Interior spaces and corridors must also be designed to allow easy movement in wheelchairs. This also includes the provision of safe elevators and staircases, as well as accessible sanitary facilities on every floor [4].

Through this project, the existing building is not viewed as a mere structure to be preserved, but rather as a real opportunity to reinvent interior spaces, employing modern technical capabilities to improve quality of life and reduce the environmental impact in the long term, without the need for a new construction.

1.2 *Standard Adherence*

This section of the report includes all the relevant standards that the project adheres to. The standards are listed in Table 1 below.

Table 1. Project-specific goals and requirements.

Criterion	Goal	Reference
Energy Performance	50% Reduction	BREEM 2017
Specific Fan Power	1.5 kW/(m3/s)	BBR, BFS 2011:6
Ventilation	7 l/s/person + 0,35 l/s/m2	AFS 2020:1
DGP	95% of the time with DGP below 0.35	Assignment Specific
sDA 300/50%	3 Points	LEED_v4.1
Daylight Factor	DFmedian ≥ 1 %	BBR, BFS 2011:6
Electric Lighting	7 kWh/m2/year	EN15193-1:2017
BELOK class	26°C	BELOK
	ADDITIONAL	
Primary Energy Use	<50 kWh/m2a	BBR, BFS 2011:6
UDI 300-3000	% of Fulfilment	Assignment Specific
Daylight Factor	DTM=0.7% in 95% of the room AND DT=2.1% in 50% of the room	BREEAM-SE 6.0
View Out and Sunlight Exposure	Check	EN17037:2018+A1
Daylight Factor and Illuminance	Check	EN17037:2018+A1

2. Methodology

This initiative employs a structured framework integrating architectural evaluation, digital simulation, and ecological appraisal to formulate an optimal strategy for refurbishing an existing office structure. An interconnected design philosophy served as the core methodology, considering illumination, airflow, power consumption, and occupant well-being as interdependent factors across all planning and execution stages.

Initial efforts comprised an on-site investigation at the Alnarp location to comprehend the contextual architectural and ecological setting and pinpoint deficiencies in present conditions regarding natural light, spatial arrangement, thermal regulation, and air movement. Subsequent meteorological evaluation utilized Copenhagen data to gauge effects of sun exposure, breezes, and exterior climate on structural efficiency.

Using gathered information, a digital representation was developed via engineering applications (Revit) and modeling suites (Rhino and Grasshopper with Ladybug). This representation segregated distinct thermal sections based on spatial purpose and cardinal direction, enabling accurate quantification of power usage and internal atmospheric states. Natural illumination efficacy was modeled employing visual-based simulation resources (Climate Studio) to verify adherence to LEED and BBR criteria for daylight factors and visual discomfort prevention.

Concerning air circulation mechanisms, current operations were scrutinized, and the feasibility of repurposing present conduits was determined. Computational methods examined air distribution, resistance dissipation, and air handling unit functionality. Where specifications were unmet, high-efficiency replacements featuring thermal recuperation and automated regulation technologies were suggested.

2.1 On-Site Building Scrutiny

2.1.1. *Location and Context*

The Green Innovation Park is located at Slottsvägen 1, 234 56 Alnarp, Sweden. It sits within the campus of the Swedish University of Agricultural Sciences (SLU), surrounded by institutional buildings and greenery. The site is open, with more tree coverage on the north and east sides. In contrast, the south and west sides, especially along the adjacent road, have fewer trees, allowing more direct sunlight. The building also has easy access to public roads, making it conveniently reachable. Climatically, Alnarp experiences a temperate oceanic climate, with moderate seasonal variation in temperature and daylight. This has direct implications for daylighting design and ventilation strategies, requiring a balance between natural light access and thermal comfort management.

2.1.2. *Existing Conditions*

The building is currently being renovated and was previously used as an office. It has a strong structure and an existing mechanical ventilation system that may be reused if it meets the current standards. The renovation focuses on improving daylight, ventilation, energy use, and thermal comfort, while also maintaining the original character of the building.

Main design considerations include:

- Improving indoor comfort and lowering energy use.
- Reusing existing HVAC systems where possible.
- Following Swedish building codes and ensuring accessibility for all users.

2.1.2.1. *Basement Floor*

The basement was not in active use during the site visit. It was mostly used for storage. The space had a low ceiling height, which makes it unsuitable for regular activities. There was also a crawl space underneath the main structure, likely used for technical systems. A toilet was located on the east side, and there was a ramp exit on the west side, possibly used for service or emergencies. Overall, the basement plays a minor role in the building's function unless major changes are made.

2.1.2.2. *Ground Floor*

The ground floor is the main level for access and activities. The main entrance is on the south side and has a ramp for people with physical disabilities. A secondary entrance is located on the north side. Inside, the layout includes a reception or lobby area, several offices, meeting rooms, and a large open workspace. There is also a small kitchen for staff and a multi-functional cafeteria that seems to be used as a dining area and a conference room. The space is easy to move through, but the building has no elevator, only stairs, so people with mobility challenges can only use the ground floor. The floor opens up to green outdoor areas, which adds to the comfort of the space.

2.1.2.3. *First Floor*

The first floor appears to be used for academic or research purposes. A large lecture hall with a tall ceiling is located on the south side. On the east side, there are labs or research rooms, and on the west, there are closed offices or meeting rooms. A toilet is also available on this floor. The hallway is enclosed and not well lit due to a lack of windows. There is a service room that provides access to the attic for maintenance work. Like the rest of the building, this floor can only be reached by stairs, which limits access for people with mobility issues.

2.1.3. Design Approach

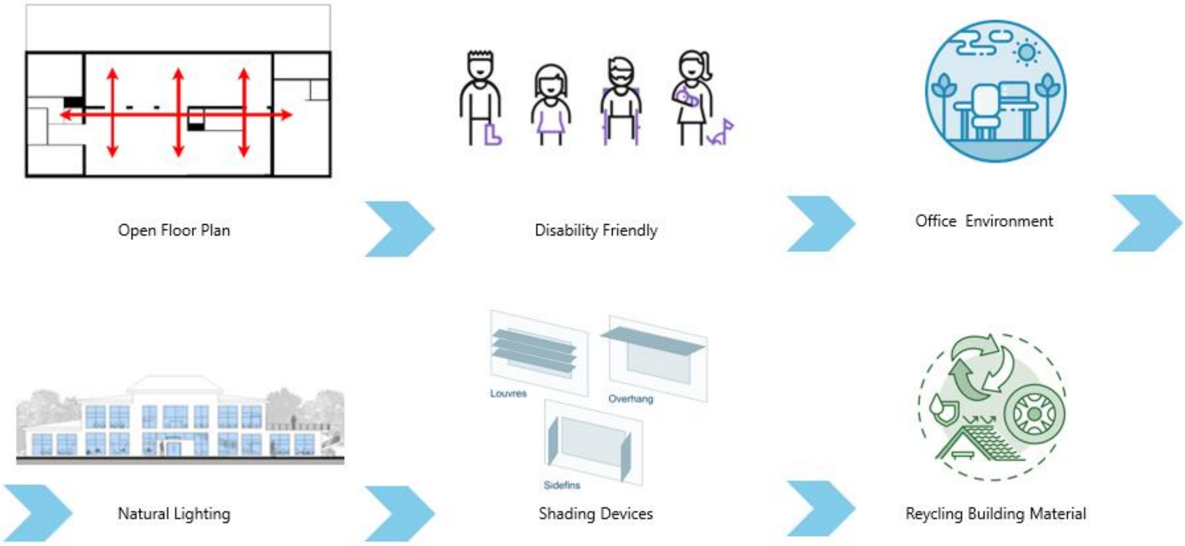

Figure 1. Design Approach to Sustainable, Accessible, and Human-Centered Building Design.

The building design adopts a sustainable, accessible, and human-centered approach, beginning with an open floor plan that promotes spatial flexibility and barrier-free movement. This layout allows for dynamic, multi-use spaces that support collaboration, adaptability, and an enhanced user experience. By removing unnecessary partitions and creating fluid zones, the design ensures ease of navigation and interaction among occupants.

Accessibility is a central consideration, with features tailored to meet the needs of individuals with varying physical abilities. The design includes wide corridors, gently sloped ramps, and universally accessible fixtures, all in alignment with Swedish and EU accessibility standards. These elements ensure that people with mobility, visual, or auditory challenges can use the space safely and independently.

The office environment has been carefully crafted to support both productivity and well-being. Ergonomic workstations are distributed throughout the space alongside quiet zones for concentration and open areas that encourage collaboration. Biophilic design principles are also integrated, using natural materials and greenery to enhance the interior atmosphere and promote mental health.

Natural lighting plays a key role in reducing energy use and improving comfort. The design maximizes daylight through strategically placed large windows, skylights, and reflective interior surfaces. This approach reduces reliance on artificial lighting while creating bright, uplifting spaces that support occupant wellness.

To enhance thermal comfort and minimize energy loads, the building incorporates shading devices such as overhangs and louvers. These elements regulate solar gain, prevent overheating, and reduce glare, contributing to a more comfortable and energy-efficient indoor environment.

Sustainability is further addressed using recycled and reclaimed building materials, including wood, duct, pipe, etc. This strategy reduces construction waste and supports circular economy principles, significantly lowering the building's carbon footprint while promoting responsible resource use.

2.1.4. *Improved Case*

2.1.4.1. Ground Floor

The design includes two expansive open office zones Figure 28, North (190 m^2) and South (234 m^2), that support high-density collaborative work. Meeting rooms (32 m^2 and 31 m^2) are placed at the periphery to allow easy access without disrupting the workflow in the open areas. The floor also features a canteen (31 m^2), kitchenette, and pantry, promoting employee well-being and providing spaces for relaxation and informal interaction. Additionally, there is a rest room (14 m^2), a server room (6 m^2), and a technical room, along with adequate storage. These facilities support both the operational and human needs of the office environment.

2.1.4.2. First Floor

The layout includes a large open office area (119 m^2) designed to encourage collaboration Figure 29, along with a head office (42 m^2) and a conference room (43 m^2) for meetings and private discussions. There are four smaller private offices, each around 15 m^2, which are ideal for focused work or management use. A central corridor (78 m^2) provides efficient circulation throughout the floor, and a sanitary area (15 m^2) includes accessible restrooms. This floor reflects BEM principles through its modular design, the balance between open and private workspaces, and the emphasis on productivity through acoustic separation and space optimization.

2.1.4.3. Accessibility and Circulation Improvements

A key transformation in the renovated design is the reconfiguration of the ground floor into a single-level, physically challenged-friendly layout, a critical element that was missing in the base case. The entire floor is designed for barrier-free access, ensuring inclusivity and compliance with universal design standards. An elevator has been introduced at the entrance, significantly improving vertical accessibility and enhancing movement between floors for all users.

Additionally, the relocation of the staircase has created a more open spatial arrangement and improved visual connectivity. This change not only facilitates intuitive navigation but also enhances the sense of openness and clarity in how spaces are experienced and accessed.

2.1.4.4. *Basement Level*

The basement floor has been retained as a dedicated storage area, preserving its original function while also offering potential for adaptive reuse. This space could be rented out as auxiliary storage, providing an opportunity for additional income generation without interfering with the primary office functions above. This approach aligns with the BEM philosophy by optimizing underutilized space for economic and operational benefit.

2.1.5. *Sustainability and Efficiency*

Throughout both floors, the layout demonstrates a strong focus on sustainability and energy efficiency. Large windows allow natural daylight to permeate work areas, reducing the need for artificial lighting. A compact vertical core minimizes the energy load from elevators and HVAC systems. Shared facilities reduce redundancy, and circulation areas are designed for efficient movement, all contributing to an energy-conscious, human-centered office design.

2.1.6. *Comparison with the Base Layout*

Compared to the base layout, which lacked communal or social areas, the renovated design includes several such amenities, fostering interaction, relaxation, and employee well-being. The inclusion of kitchenettes, sanitary blocks, and casual seating further reflects attention to occupant needs. The redesign not only improves functional performance and user comfort but also aligns with inclusive, sustainable, and modern workplace design principles.

2.1.3 *Climate and Site Analysis*

Due to the lack of accurate climate data specific to the Alnarp location, data from Copenhagen was used, as it is geographically closer and has similar climatic conditions. This data indicates that the region has a cold and humid climate, with temperatures ranging from -9.2°C to 30.4°C, and relative humidity ranging from 27% to 100%, with the highest values recorded during the winter, can be seen in Table 2. Analysis also shows heavy cloud cover throughout most of the year, which negatively impacts daylight and increases the need for efficient ventilation and heating solutions.

Table 2. Climate data extracted from Climate Studio

Parameter	Climate Studio Output
Outdoor temperature range (°C)	-9.2 to 30.4
Relative humidity range (%)	27% to 100%

Analysing solar radiation reveals that the building's various facades are exposed to varying levels of radiation throughout the year. The south-west facade recorded the highest radiation levels, averaging 854 W/m^2, followed by the southern facade. The north-east facade receives the lowest radiation, mostly under 300 W/m^2. This variation highlights the importance of orienting heavily used spaces toward the south and southwest to maximize solar energy and daylight during the cooler seasons, while also ensuring shading during the summer.

In addition, the results of the visual site survey showed that the northern and eastern facades suffer from heavy shading from tall trees and adjacent buildings, reducing the quality of daylight in those directions and making them less effective in providing the spaces with beneficial radiation. Wind direction analysis also showed that the prevailing winds emanate primarily from the southwest and west, opening the door to effective natural ventilation solutions into the orientation of windows and openings in future designs to enhance cross-ventilation and reduce reliance on mechanical systems.

2.2. Daylight Performance

The building's daylight performance was evaluated through a series of simulations using the Honeybee tools within Grasshopper. Only the load-bearing structure was retained in the digital model to ensure the analysis focused on the actual distribution of light within the spaces. The simulations adhered to international standards such as LEED and BREEAM-SE, along with local engineering considerations.

The simulations included key indicators including Daylight Factor (DF), Spatial Daylight Autonomy (sDA), Daylight Glare Probability (DGP), Useful Daylight Illuminance (UDI), and Direct Sunlight Exposure (DSE).

2.2.1. *Daylight Factor (DF)*

The results showed that the DF median was 0.65% on the ground floor and 0.44% on the first floor, DF mean was 0.98% on the ground floor and 0.85% on the first floor, values well below the minimum requirement (1%)

according to the BBR and LEED standards. The uniformity ratio was 9.5% on the ground floor and 4.5% on the first floor, indicating poor uniformity in light distribution, can be seen in Table 3 & appendix G.

According to the BREEAM-SE standard, 95% of the area must have a DF ≥ 0.7%, and 50% of the area must have a DF ≥ 2.1%. However, as of now, only 12% of the areas on the ground floor and 13% on the first floor achieved a DF ≥ 2.1%, results that indicate substandard performance, can be seen in Table 3.

2.2.2. *Spatial Daylight Autonomy (sDA300/50%)*

The simulation recorded very low levels of light independent of artificial sources, with 9.46% on the ground floor and 7.98% on the first-floor can be seen in appendix G, compared to the target of 75% of occupied areas according to LEED requirements. This deficiency is primarily attributed to dense shade from surrounding trees, poor light transmittance (T-vis) of the glass, and a low window-to-wall ratio (WWR).

2.2.3. *Daylight Glare Probability (DGP)*

A DGP analysis showed that 12.85% of the office space on the ground floor and 7.21% on the first floor can be seen in appendix G suffered from unacceptable glare, can be seen in Table 3, levels well above the maximum permissible limit of 5% of the office space during working hours, according to international standards.

2.2.4. *Useful Daylight Illuminance (UDI)*

Within the ideal illumination range (between 300 and 3,000 lux), the achieved values were 30.3% on the ground floor, and 21.64% on the first floor, can be seen in appendix G. These values remain suboptimal and demonstrate the current system's limited ability to provide sustainable, useful daylight, can be seen in Table 3.

2.2.5. *Direct Sunlight Exposure (DSE)*

The analysis showed that spaces on the ground floor received an average of 15.64 hours of direct sunlight between February 1 and March 21, while this figure increased to 26.65 hours on the first floor, can be seen in appendix G. However, these figures alone are not sufficient to meet the requirements for continuous or balanced exposure to daylight.

Table 3. Daylighting results in the base case

	Ground Floor	First Floor
DF median	0.65%	0.44%
DF mean	0.98%	0.85%
Uniformity ratio	9.5%	4.5%
DTM (BREEM-SE)	0.7 in 45% of space 2.1 in 12% of space	0.7 in 31% of space 2.1 in 13% of space
sDA	9.46%	7.98%
DGP	12.85%	7.21%
UDI	30.3%	21.64%
Direct Sun Exposure	Average of 15.64 hours	26.65 hours

Based on the simulation results, it was found that the building's daylighting performance does not meet the desired requirements in terms of efficiency and visual comfort, caused mainly by intensive shading by foliage around the building, insufficient T-vis values of windows, and insufficient WWR for the location. Requiring thoughtful design improvements. One of the most important proposals that could contribute to improving performance is the use of glass with a visible light transmittance (T-vis), as a balanced option that allows sufficient daylight without causing excessive glare, especially in areas directly exposed to solar radiation.

Regarding the thermal properties of windows, the study showed that the current heat transfer coefficient (U-value) values range between 1.13 and 1.3 $W/m^2 \cdot K$, which are relatively good values but can be improved to reduce heat loss and improve the building's thermal performance during the winter. It is also recommended to expand window openings to maximize daylighting during daylight hours.

2.3. Electric Lighting

The electric lighting system's performance was simulated in Dialux Evo 13.0 (Figure 24, Figure 25, Figure 26, Figure 27) adhering to EN 15193 for calculating the Lighting Energy Numeric Indicator (LENI) expressed in $kWh/(m^2 \cdot a)$. Target illuminance levels were set at 300–500 lux for work areas and 100–200 lux for ancillary zones, following functional zoning. Luminaires were positioned to balance task and ambient lighting, with operational schedules of 3 650 hours per year defining runtime. Energy demand was calculated using fixture efficacy and system control parameters, while daylight autonomy (DA) and artificial lighting demand (LD) outputs from simulations informed the LENI aggregation across room types.

2.4. Energy Performance

Tackling energy solutions for this project is challenging and requires extensive fine-tuning and assumptions, primarily due to the building's unique nature, age, and limited on-site time with the building itself.

2.4.1. *Existing Building*

The scrutiny of existing energy performance is subject to many assumptions regarding the building's structure, the performance of each component, occupancy, and internal loads. The overall goal, however, is to achieve EUI levels in the simulations that are similar to those of the existing building, retrieved from the energy declaration, which serves as a starting point. The performance of assumed existing components, along with the material composition, can be seen in Table 4.

Table 4 Existing components of the building

	Components		
	Roof	**Façade**	**Ground Floor**
U-value [W/(m2*K)]	0.34	2.096	2.94
Composition Exterior to interior	Cellulose 100mm Concrete 200mm	Brick 400mm	Concrete 400mm

Additionally, the building is split into two energy zones in Climate Studio to reflect the standing conditions better, and corresponding zone settings can be seen in Table 5.

Table 5. Zone settings for the existing building[5]

	Main Building		Basement	
	Value	schedule	value	schedule
People [p/m²][1]	0.1	Flexible Hours	0.1	1h every workday
Eqipment [W/m²]	15	Flexible Hours	2	1h every workday
Lights [W/m²]	10	Flexible Hours	10	1h every workday
Hot Water [m3/h/p]	0.001	Flexible Hours	off	
Heating Setpoint [°C]	21		15	
Cooling Setpoint [°C]	25		off	
Airflow /person [l/s]	7		0	
Airflow /m2 [l/s]	0.35		0.35	
Heating COP	1		1	

[1] BBR, 2020

2.4.2. Renovated Building

Based on the assessment of the building on site, the existing structure did not feature any insulating material on the façade and ground floor. Roof, however, has been recently renovated and insulated with cellulose. Therefore, only uninsulated components were subject to renovation further to lower the heating energy demand of the building. The final state of the components can be seen in Table 6.

Table 6. Renovated building components.

	Components		
	Roof	Façade	Ground Floor
U-value [W/(m2*K)]	0.34	0.15	0.54
Composition Exterior to interior	Cellulose 100mm Concrete 200mm	Mineral wool 200mm Brick 400mm	Concrete 400mm Mineral wool 50mm

The energy performance is being simulated in "Climate Studio", where different zones of the building were assigned, different values based on their assumed occupancy, internal loads, and other factors affecting the energy performance of the building, and can be seen in Table 7.

Table 7. Energy simulation inputs for different energy zones

	Building energy zones						
	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7
People [p/m²]	0.01	0.56	0.1	0.23	0.01	0.01	0.04
Eqipment [W/m²]	15	15	15	10	12	Off	15
Lights [W/m²]	10	10	10	10	10	10	10
Hot Water [m3/h/p]	Off	0.001	0.001	Off	Off	0.001	Off
Heating Setpoint [°C]	21	21	21	21	21	21	21
Cooling Setpoint [°C]	OFF	26	26	26	26	26	26
Airflow /person [l/s]	0	7	7	7	0	7	7
Airflow /m2 [l/s]	0	0.35	0.35	0.35	0.35	0.35	0.35
Heating COP	Off	1	1	1	1	1	1

Table 8. Energy zone room composition

	Room1	Room2	Room3	Combined Size [m²]
Zone 1	Elevator Shaft	Technical Shaft		23.2
Zone 2	Canteen			88.8
Zone 3	Ground Floor Open Office	First Floor Open Office		672
Zone 4	Meeting Room 1	Meeting Room 2	Conference Room	128
Zone 5	Server Room	Technical Room		60.7
Zone 6	Ground Floor Sanitary	First Floor Sanitary		55
Zone 7	Rest Room	Storage Room	Pantry	24.66
Zone 8				

Moreover, to scrutinise the energy performance of the building, the BBR EP_{pet} equation is used [5], where all the necessary values are retrieved from "Climate Studio" energy model simulations (Equation 1)

$$EP_{pet} = \frac{\sum_{i=1}^{6}\left(\frac{E_{uppv,i}}{F_{geo}}+E_{kyl,i}+E_{tvv,i}+E_{f,i}\right)\times PE_i}{A_{temp}} \quad (1)$$

PE_i value from the equation is set at 0.9, as per BBR standard, due to the building being located in Skåne, a Swedish region[5].

The HVAC section of the report includes the BELOK class, along with specific fan power methodology and results. Moreover, all the schedules used for different zones will be mentioned in the appendix A section of the report.

To ensure the simulations from Climate Studio meet the highest standards, complex tree shading and a seasonal schedule are implemented into the model. The context around it is also modelled with the same goal. Moreover, the shading located on the building itself is discussed in the daylighting section of the report.

2.5. HVAC

The HVAC system in the building consists of two existing air handling units (AHUs), one used for conditioning the ground floor of the building and the second AHU used to condition the first floor. This same approach is also used in renovated buildings to improve the economic viability of the renovation, theoretically. Thus, it is crucial to reuse as much of the existing system as possible. This reuse method of renovation extends to the ducting system, where a certain percentage of the ventilation ducts are repositioned to accommodate the new building layout.

2.5.1. *Existing HVAC system*

The two existing Air Handling Units on both floors of the building are scrutinised based on the airflow capabilities of the current system. This step ensures the compatibility of the system with the newer hygienic needs of the building, as per current standards seen in the Table 9. All necessary information about the current system's capabilities is sourced from existing drawings of the system, where each building zone has a specific supply and exhaust airflow noted. Furthermore, with this information, the exhaust and supply diffusers are categorised according to their capabilities for later reuse in the new system.

Moreover, the ducting of the existing system is modelled in 3D CAD software Revit, where, similarly to the previous paragraph, the ducting is categorised based on its different diameters and lengths. The lengths of the ducts are multiplied by 0.8, or 80%, as shown in Equation 2, to account for scrap that cannot be reused in the new system.

$$PRL = L \times 0.8 \tag{2}$$

$PRL \ldots$ *Possible Reuse Lenght*
$L \quad \ldots$ *Existing Duct Lenght*

2.5.2. *Renovated HVAC System Design*

The Ventilation system is used for hygienic air supply during the heating season and hygienic and thermal conditioning during the cooling season. To ensure the building achieves compliance with national standards, as well as the overheating criteria chosen for the building, each building zone must be scrutinised separately for both scenarios. This means that the zone has either higher airflow needs due to internal thermal loads or standard airflow following the regulations.

Equation 3 calculates the airflow in accordance with national standards, and Equation 4 calculates the airflow during the overheating period. Each calculation is performed on each zone separately, and the higher value is chosen which then adds up to the required airflow need of the AHU units.

$$L = (0.35 \times A) + (7 \times P) \tag{3}$$

$L \ldots Air\ flow\ [l/s]$
$L \quad \ldots Total\ area\ of\ zone\ (m^2)$
$P \quad \ldots Number\ of\ occupants$

$$L = \frac{Q}{\rho * c * (T_{exhaust} - T_{supply})} \tag{4}$$

$L \ldots\ Air\ flow\ [l/s]$
$Q \ldots Energy\ flow; "Peak\ Cooling\ demand"$ [J/s]
$\rho \ldots Density\ of\ air \sim 1.2\ [kg/m^3]$
$c \ldots Specific\ heat\ capacity\ of\ air \sim 1.000\ [J/kgK]$
$T_{exhaust} \ldots Temperature\ of\ exhaust\ air; Max\ overheating\ temp.\ [°C]$
$T_{supply} \ldots Temperature\ of\ supply\ air$ [°C]

The peak cooling demand is extracted from a dynamic energy simulation in "Climate Studio". To overcome thermal inertia problems, the cooling system's setpoint is set one degree below the overheating criteria. Heat recovery is turned on with an efficiency of 70% for both AHUs.

The overheating criteria according to BELOK classes are chosen to be 26 °C[6]. This temperature class is selected due to the strict energy requirements of the assignment, lowering the cooling energy need. Moreover, as a temporal criterion, FEBY 18's limit of 10h-% overheating within the standard months of April to September was used instead of BELOK's maximum 80 work hours a year [6], [7].

Table 9. Settings for the ventilation system of the energy model. [1], [3]

Min. Airflow[1,2] [l/m²s]	Additional Airflow[1] [l/sPerson]	Min. Total Flow Occupancy [l/s]	Cooling Setpoint [°C]	Overheating Criteria[1] [°C]	Supply Air Temp. Cooling [°C]
0.35	7	953	25	26	18
[1] BELOK, 2015 [2] BBR, 2020					

Moreover, the ventilation system is designed to handle the internal loads of up to 80 people in open office areas with a corresponding schedule (Figure 2).

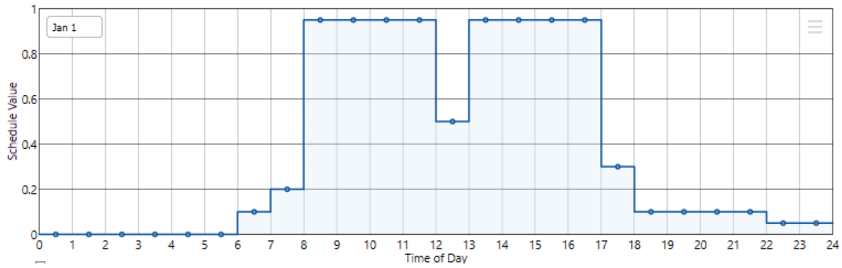

Figure 2. OfficeMedium BLDG_OCC_SCH schedule for internal loads in Climate Studio

Furthermore, while designing the duct system, the overall goal is to reuse existing components to a certain extent. This project aimed for an average friction loss of 1 Pa/m throughout the duct system. The ASHRAE friction chart was used to determine the duct sizes.

The position and dimension of heating and cooling coils and overall air treatment using different AHU modules, such as filters and dampers, are calculated using the Mollier Chart. It is necessary to use this diagram to properly analyse and visualise how different AHU components will affect the air conditioning from outside to the inside, according to the exterior weather conditions in the Copenhagen area. The exterior weather conditions used as input are shown in Table 10 [8].

Table 10. Critical points for sizing heating and cooling coils [8]

	Temp.[1] [°C]	[%]	Date[1]
Min dBT	-9.2	77	06. Jan 05:00
Max dBT	30.4	48	21. 8 16:00
[1] climate.onebuilding.org			

Moreover, to ensure adequate power requirements for the existing heating coils to be adapted into the renovated building and the dimensioning of new cooling coils, values obtained from the Mollier chart are calculated using equation 5. This step is necessary to determine how much power the heating and cooling coil will need to condition the air to the desired temperature suitable for an office space.

$$P = V \times \rho \times \Delta h \tag{5}$$

$P \ldots Power$ [kW]

$V \ldots\ Airflow\ rate\ [m^3/s]$

$\rho \ldots\ Air\ density$ [kg/m^3]

$\Delta h \ldots Change\ in\ Enthalpy$ [kJ/kg]

Additionally, equations 6 and 7 calculate a fan's electric power and specific fan power (SFP).

$$\dot{W}_t = \frac{\dot{V} \times \Delta p_{tot}}{\eta_{tot}} \tag{6}$$

$\dot{W}_t \ldots Electrtic\ power\ to\ fan$ [kW]

$\dot{V} \ldots Airflow\ rate\ trough\ fan\ [m^3/s]$

$\Delta p_{tot} \ldots Total\ pressure\ rise\ of\ the\ fan$ [KPa]

$\eta_{tot} \ldots Total\ efficiency\ of\ the\ fan\ [-]$

$$SFP = \frac{\dot{W}_t}{\dot{V}} \tag{7}$$

SFP … $Specific\ fan\ power$ $[\mathrm{kW}/(m^3/\mathrm{s})]$

Furthermore, the overall layout of the supply and extract diffusers is designed in SWEGON “RUD” or Room Unit Designer [9]. This approach ensures functional layout distribution of the different diffusers while providing important data concerning throw lengths, drafts in different spaces, and sound levels inside the building.

Table 11. Air Terminal Units Design Standards[5], [10]

Air Velocity[1] [m/s]		CO2 Content[2] [ppm]	Maximum Pressure Drop [Pa]	Noise Level [dB]
Summer	Winter			
≤0.25	≤0.15	<1000	≤50	≤30
[1]BBR, 2020 [2]AFS 2009:2 Section16				

2.6. Heating System

The hydronic heating system was developed through coordinated BIM modeling in Revit Figure 21 and detailed thermal analysis in Climate Studio, with radiator sizing performed via Effektsimulering-Integra40-Plan to ensure emitter capacities matched room demands. Initial load calculations determined baseline heat requirements, to which a 20% safety factor was applied to accommodate occupancy variations, infiltration rates, and temperature fluctuations. To counteract thermal stratification, 10% of the upper-floor load was shifted to the lower-floor loop, promoting balanced vertical temperature distribution[11]. Hydraulic critical path analysis employed the Friction Method with a Hazen-Williams coefficient for emitter piping and accounted for both major and minor losses, constrained by a design friction limit. Pump selection was guided by desired flow velocities within turbulent yet non-erosive regimes and calculated dynamic head requirements using pump affinity laws. The piping layout centralized vertical risers in the plant room, utilized ceiling-mounted headers for horizontal distribution, and integrated hydraulic balancing valves at terminal branches to achieve uniform flow. Copper piping was specified, noting its thermal expansion coefficient of 17×10^{-6} /°C Table 17, and routing decisions were validated against structural and aesthetic constraints in the BIM model.

3. Results

4.1. Daylight Performance

Evaluating Daylighting Performance After Renovation.

During the renovation phase, the project sought to improve the quality of daylighting within the Green Innovation Park building through an architectural design based on approved European and international standards, specifically the requirements of Boverkets Byggregler (BBR), LEED v4.1, and European Standard EN 17037:2018+A1:2021. This was achieved through a series of design improvements, including redistributing openings and increasing the size of windows on the north and south facades, in addition to adding a skylight on the first floor to enhance the level of daylighting in central areas that are difficult to access, which contributed to efficiently increasing daylight penetration to meet the daylighting needs of the office spaces.

To achieve a visual and thermal balance, high-performance glass was used with a visible transmittance (T-vis) of 0.713, a solar heat gain coefficient (SHGC) of 0.605, and a low thermal transmission value (U-value) of 0.9 $W/m^2 \cdot K$, allowing sufficient light to pass through without excessive heat gain. Multiple shading systems were also adopted to reduce glare: on the south façade, external vertical columns were used alongside dynamic blinds, while the windows on the east and west sides of the first floor were equipped with dynamic blinds only.

A pivotal strategy in the new design was the reconfiguration of the ground floor into an open-plan workspace, which helped improve the uniform distribution of daylight and increase its efficiency within the work environment. These design interventions underscore the project's commitment to achieving visual comfort and sustainability standards, in accordance with the guidelines of the EN 17037 standard, particularly with regard to providing adequate levels of natural light without causing visual or thermal disturbance.

After conducting a series of lighting simulations to ensure that the building renovation complies with regulatory requirements and international environmental standards, including Boverkets Byggregler (BBR), BREEAM-SE, LEED v4.1, and the European standard EN 17037:2018+A1, the simulation results demonstrated high performance levels, demonstrating the effectiveness of the applied design solutions, can be seen in Table 12.

Achieving Daylight Factor (DF) Requirements: BBR regulations stipulate that office spaces must achieve either a DFpoint ≥ 1% or a DFmedian ≥ 1%. The BREEAM-SE standard also requires that 95% of the area achieve a DF ≥ 0.7%, and 50% of the area achieve a DF ≥ 2.1%. After implementing the design modifications, the simulation results recorded: The average daylight coefficient after renovation (DFmedian = 2.9%) and (DFmean = 4.7%). 93% of the area achieved a DF of ≥ 0.7%, while 64% of the area achieved a DF of ≥ 2.1%. The homogeneity rate reached 11.58%, can be seen in Table 12 and appendix H.

The Spatial Daylight Autonomy (sDA300/50%) reached 75%, which meets the LEED requirements for maximum points in this area. The UDI ratio within the ideal range (300–3000 lux) reached 84%, can be seen in appendix H reflecting the provision of comfortable and effective levels of natural light within the occupied spaces.

Regarding Daylight Glare Probability, the simulation showed that the percentage of the area affected by DGP levels exceeded the permissible limits in only 4.3% of the area, can be seen in appendix H, indicating that the design was able to maintain glare levels within the limits recommended by the European standard EN 17037.

Direct Sunlight Exposure: A dedicated simulation was conducted to analyse the duration of direct sunlight exposure within the occupied spaces during the period from February 1 to March 21. The values were based on Sweden Council's criteria:

Minimum	**Medium**	**High**
1.5 hours per day	3 hours per day	4 hours per day

In the improved case, the average sun exposure during this period was 116.9 hours on the ground floor, which is 2.39 hours per day, and on the first floor 113.14 hours, which is 2.31 hours per day, can be seen in appendix H. These values confirm that the minimum performance requirements were fully met, approaching the average level, demonstrating the design's effectiveness in enhancing access to natural sunlight without causing visual disturbance.

Table 12 Daylight Performance Results After Renovation

Indicator	Measured Value	Standard Requirement	Compliant?
DFmedian	4.7%	DFpoint ≥ 1% or a DFmedian ≥ 1%	Yes
DFmean	2.9%		
Area with DF ≥ 0.7%	93%	≥ 95% (BREEAM)	No (very close)
Area with DF ≥ 2.1%	64%	≥ 50% (BREEAM)	Yes
Uniformity Ratio	11.58%	—	Yes
sDA300/50%	75%	≥ 75% (LEED v4.1)	Yes
UDI (300–3000 lux)	84%	Within range (EN 17037)	Yes
DGP > acceptable limit (5%)	4.3%	≤ 5% (EN 17037)	Yes
Direct Sunlight Exposure – Ground Floor	2.39 h/day	≥ 1.5 h/day (Stockholm – EN 17037)	Yes
Direct Sunlight Exposure – First Floor	2.31 h/day	≥ 1.5 h/day (Stockholm – EN 17037)	Yes

4.2. View Out Based on European Standard EN17037:2018+A1

An analysis of the quality of view out from the interior spaces was conducted using an illustrative vertical section (see Figure 3), to verify the architectural design's compliance with the requirements of the European Standard EN 17037. The standard stipulates that glazed openings must provide a clear view of three horizontal layers: the sky, natural or urban elements (landscapes or buildings), and the ground, from a vantage point representing the user's position within the space.

In this analysis, an eye height of approximately 1.2 meters above ground level was adopted, in line with the reference values specified in the standard. The visual representation results show that the design provides an uninterrupted view of the three required layers of seating on both floors, demonstrating the achievement of a "high view out" level according to the standard's qualitative classification. This reflects the success of the architectural treatment of the windows, in terms of height, location, and viewing angle, in providing a comfortable and direct visual connection with the outdoor environment.

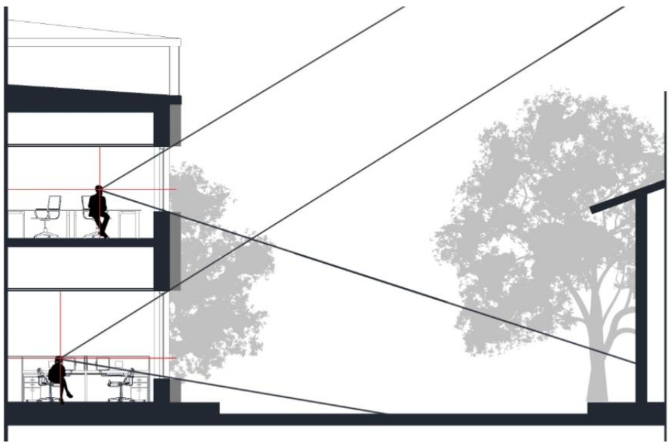

Figure 3. View Out Analysis for Ground and First Floor Users

4.3. Electric Lighting Analysis

Table 13. Results for PV Panel Production

PARAMETER	VALUE	DETAILS
Maximum Energy Demand	5772 kWh/a	Peak annual energy consumption for lighting.
Estimated Energy Demand	4944 – 5772 kWh/a	Expected annual energy range, reflecting operational variability.
Maximum Energy Saving	828 kWh/a (14%)	Potential savings achievable through efficiency measures.
LENI	5 kWh/(m^2 * a)	Lighting Energy Numeric Indicator, indicating overall energy efficiency.
CO_2 Emissions	1983 – 2315 kg/a	The annual carbon footprint associated with lighting energy use.[SR1] [SR2]

The total annual lighting energy demand was determined to range between 4 944 kWh Table 20 and 5 772 kWh Table 21. Applying optimized control strategies and potential retrofits yielded a maximum savings of 828 kWh/a (14%). The calculated LENI met the benchmark of 5 kWh/(m^2·a), confirming compliance with efficiency targets. CO_2 emissions were estimated at 1 983–2 315 kg per annum, directly proportional to energy use. Highest energy intensities occurred in kitchens and open-plan offices, indicating priority areas for further efficiency measures. Detailed room-specific LENI values and illuminance distributions are provided in the supplementary documentation.

4.4. Energy Performance

The renovated building's final energy performance, the final EPpet value, is 48.132kWh/m^2. This value is achieved by calculating Equation 1 in the methodology section 2.4.2. Individual values used in the equations are as follows in Table 14.

Table 14. Individual energy use of different building sources.

Heating Energy Use [kWh/m²]	34.88
Cooling Energy Use [kWh/m²]	0.708
Artificial Lighting Energy Use [kWh/m²]	5
Hot Water Energy Use [kWh/m²]	3.001
AHU Fan Energy Use [kWh/m²]	8.823
Property Energy Use [kWh/m²]	1.069

As the Table 14 shows, the highest energy use in the building comes from heating the individual building zones, while cooling uses a marginal amount. This is mainly due to the extensive shading provided by greenery such as trees and other contextual buildings. Another reason is that the building lacks insulation from the ground soil underneath. With an additional 50 millimeters of insulation introduced to the top of the ground floor, the overall heating energy use decreased by ~15-17 kWh/m^2.

Additionally, as discussed in the methodology, this project adheres to the 26°C BELOK classification[6] and FEBY 10%, April-September overheating criterion[7]. After renovation, this project can maintain a temperature of under 26°C for almost the entire analysis period, with an overheating percentage of only 1.042%. Again, this can be attributed to the fact that the building is situated between densely green areas with numerous trees surrounding it, which shade it from direct radiation and prevent overheating.

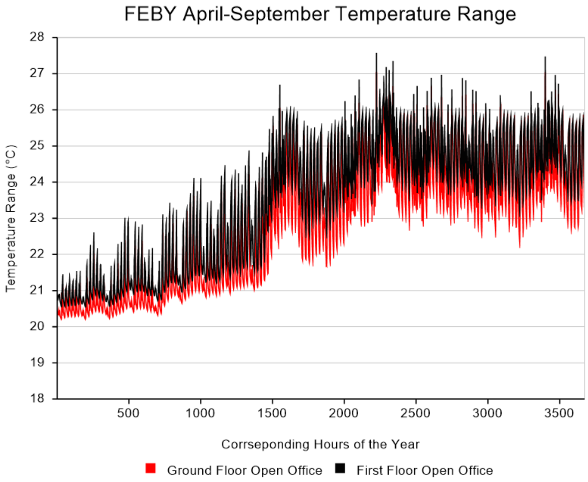

Figure 4. FEBY overheating standard period [12]

(Figure 4) This chart highlights the overheating criteria period set by FEBY from April to September. On the X-axis, the corresponding hours of the year are noted, while on the Y-axis, temperature ranges in two scrutinised zones are shown. As these are the warmest zones during the period, due to the highest occupancy rate, equipment load and overall internal thermal loads, the graphs of the other zones are not shown in the report, as they fall below the 26°C mark.

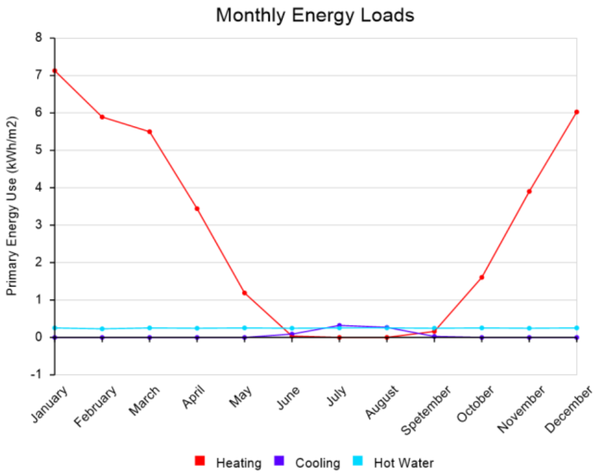

Figure 5. Monthly Energy Load

(Figure 5) The chart highlights the energy use of each load separately. The building's heating energy use is highlighted in red, which vastly dwarfs the other energy loads. Cooling is used only during the summer or warmer periods of the year. Hot water, however, stays constant throughout the year.

4.4.1. *Comparison Between the Base Case and The Improved Case*

Figure 6 seen below, compares different values used for calculating the EP_{pet} equation. A clear difference can be seen between the heating energy demand, which is significantly higher in the base case. Moreover, AHU fan power could not be included in the direct comparison due to modelling challenges in ClimateStudio, trying to achieve the same airflows as seen in the existing building drawings, so the decision was made to leave it out of the following figure.

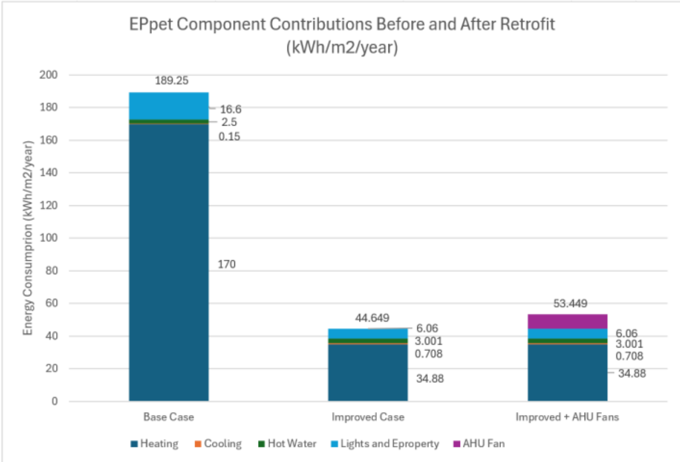

Figure 6. Different EPpet equation values comparison

Furthermore, Figure 7 highlights the difference between the peak energy consumption of cooling and heating in both cases. Again, as expected from the previous figure, peak heating load is much larger in the base case scenario compared to the improved case. As the ground floor is assumed not to be insulated in the base case, it was found that most of the heat escapes through the ground due to the relatively low temperature most of the year. The cooling peak load remains relatively low in both cases due to the extensive greenery shading that encompasses the building.

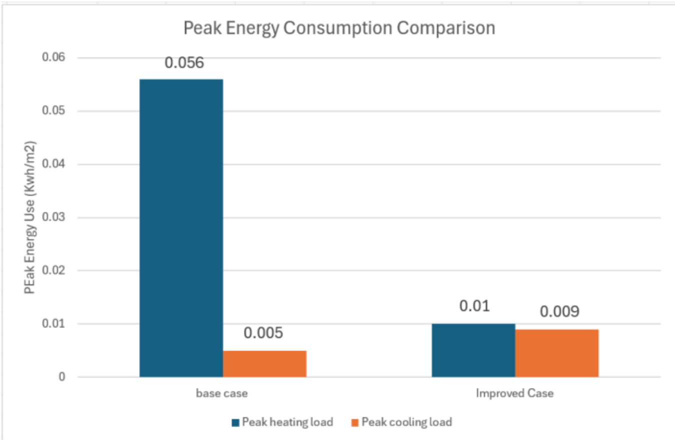

Figure 7. Peak Energy Consumption Comparison

Just how much the additional insulation on the ground floor and façade helped can be seen in Figure 8. While the U-values of the renovated components of the building are higher than standards call for, BBR also states that if the EP_{pet} comes out to be below 80 kWh/m^2/year, and the mean U-value or U_m reaches no higher than 0.6 W/m^2*K, components do not need to adhere to standard of specific U-values or U_i [5]. The figure highlights the significant impact that insulating the building has on its heating energy demand. Moreover, Figure 9 highlights the difference between U_m of base case building and the improved building.

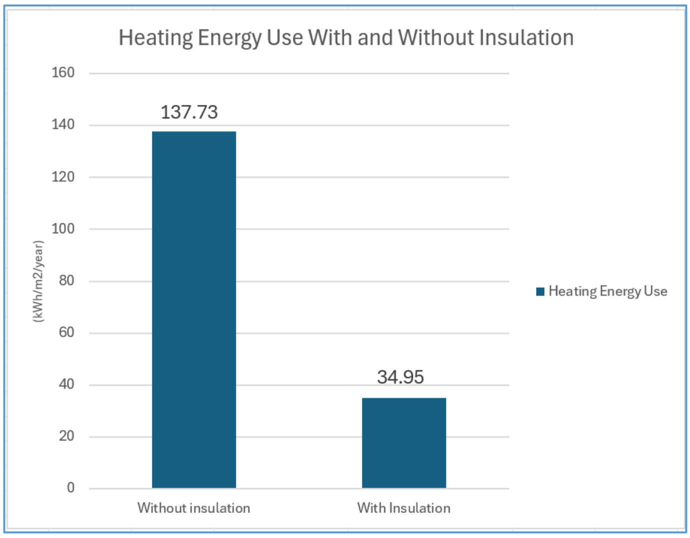

Figure 8. Heating Energy Use With and Without Insulation

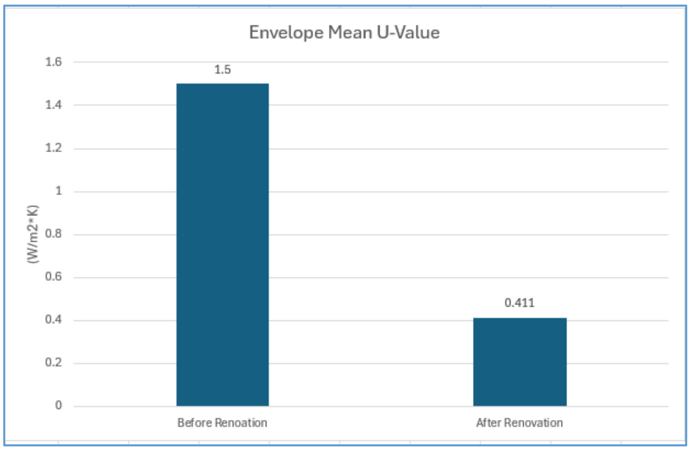

Figure 9. Mean U-Value comparison between Base Case and Improved Case

4.5. HVAC

4.5.1. *AHUs*

The current building ventilation system features two AHUs, both of which are reused in the renovated building due to their prior overhead implementations. This means that both AHUs can provide more airflow than is needed for the whole building. The two AHUs, as mentioned, are each responsible for one floor of the building.

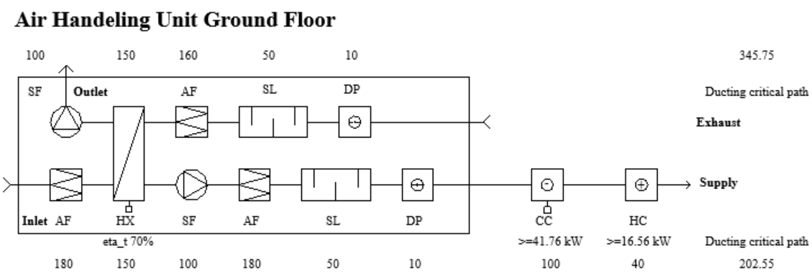

Figure 10. Ground Floor Air Handling Unit Schematics

The current AHU system, before renovation, can deliver 1560 l/s of air, while the new, renovated system only requires 1225 l/s, meaning that the current overhead for future expansion of the system is roughly 21.5%. Moreover, the current system has a heating coil connected to it as an external unit with a capacity of 10 kW, which is not enough for the new requirement of 16.56 kW, as seen in Figure 10. Meaning that the current heating coil will be moved to the upper, first-floor AHU, while this unit will have a new one installed with the required capacity. Additionally, a new cooling coil with a capacity of 41.76 kW will be added to this AHU for additional air conditioning required to fulfil new building needs. Pressure drops used for calculating SFP can be seen in Table 15.

The power of the supply run electric fan is calculated to be 1.45 kW, and the specific fan power is 1.18 kW/(m3/s). Similarly, the exhaust run power is 1.43 kW and 1.17 kW/(m3/s), respectively. The combined SFP for both the exhaust and supply fan equals 1.175 kW/(m^3/s).

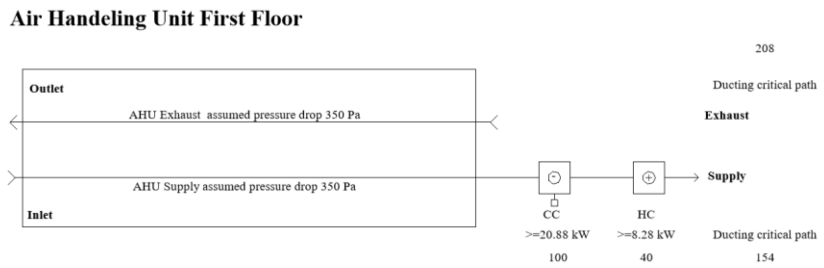

Figure 11. First Floor Air Handling Unit Schematics

Similarly, the first-floor AHU will be reused from the current system to the renovated building. It can deliver up to 620 l/s, while the new, renovated system only requires 599 l/s, meaning that the overhead is roughly 3.4%. Moreover, as previously stated, this AHU will receive the heating coil taken from the bottom, ground floor AHU, as the renovated required heating coil size is 8.28 kW, which is within the 10 kW of power capacity of the existing heating coil. Additionally, this AHU will receive a new cooling coil with a capacity of 20.88 kW to fulfil the new building needs.

The power of the supply run electric fan is calculated to be 0.54 kW, and the specific fan power is 0.9 kW/(m3/s). Similarly, the exhaust run power is 0.47 kW and 0.78 kW/(m3/s), respectively. The combined SFP for both the exhaust and supply fan equals 0.84 kW/(m^3/s).

Table 15. Complete Pressure Drops of Ventilation Parts

	Ground Floor AHU		First Floor AHU	
	Supply	**Exhaust**	**Supply**	**Exhaust**
AHU [Pa]	490	470	350	350
Heating and Cooling Coils [Pa]	140		140	
Critical Path [Pa]	202.55	345.75	154	208
Total [Pa]	**832.55**	**815.75**	**644**	**558**

APPENDIX B contains calculations for sizing heating and cooling coils for both AHUs, performed using the Mollier Diagram.

Moreover, airflow visualisations from the Swrgons Room Unit Designer can be seen in APPENDIX D.

4.5.2. Extent Of Reused Components

As specified in the HVAC methodology section, the overall goal is to reuse as many components as possible from the current system. Table 16 specifies the total extent of system reuse after the building's renovation. The quantities are extracted from the 3D model and multiplied by 0.8, as discussed in the methodology part, to account for scrap and mistakes during disassembly and subsequent assembly of the system.

Table 16. HVAC Components Reuse List

DUCTING					
Diameter [mm]	**Total Length [mm]**	**Total Reusable Length (*0.8) [mm]**	**Total Length In New System [mm]**	**Total Lenght of New Ducts [mm]**	**Reuse Percentage Out of Reusable [%]**
100	48980	39184	55938	16754	70.0
125	128925	103140	16435	0	100.0
160	43946	35156.8	21104	0	100.0
200	54855	43884	101482	57598	43.2
250	13724	10979.2	49323	38343.8	22.3
315	11743	9394.4	35566	26171.6	26.4
400	36225	28980	37644	8664	77.0
500	21943	17554.4	29477	11922.6	59.6
800x400	4610	3688	2892	0	100.0
1000x300	9369	7495.2	9369	1873.8	80.0
1000x400	3535	2828	3535	707	80.0
1000x600	2269	1815.2	2269	453.8	80.0
1200x600	9257	7405.6	1937	0	100.0
				Average Duct Reuse [%] :	**78.21**
AIR TERMINALS					
Total Number of Diffusers in Old System [Quantity]	**Total Number of Diffusers in New System [Quantity]**	**Total Number of Reused Diffusers [Quantity]**	**Total Number of Scrapped Diffusers [Quantity]**	**Reuse Percentage Out of Reusable [%]**	
72	65	39	33	**60**	

The total number of ducts reused is roughly 78%, and air terminals are reused at 60%, contributing to this project's sustainability goals. Unfortunately, due to the time constraints of the project, the reuse of fittings could not be performed due to the sheer number of them present in the current and renovated building

4.5.3. New Air Terminals

Furthermore, new supply air terminals were chosen to be "due to their quiet operation below 20 dB under project-specific airflow requirements in different zones (Figure 12). Moreover, extraction air terminals were selected based on the same criteria as the supply air terminals (Figure 13). [9]

Figure 12. Colibri Fb, Eagle CRa_low Supply Air Terminal[9]

Figure 13. Swegon CDDb -f + ALSd 200-250 Extraction Air Terminal[9]

4.6. Heating System

Total Heating Load (after 20% safety)	20.38 kW (13.42 kW Ground + 6.96 kW First Floor)
	10% of upper floor load added to ground floor
Nominal Flow Rate	1.151 L/s
	1.2 m/s
Measured Pressure Drop – Supply	4 317 Pa
	4 789 Pa
Total System Resistance	9 106 Pa
	0.93 m
Friction Limit (for Pipe Sizing)	150 Pa/m
	Copper
Thermal Expansion Coefficient	17×10^{-6} /°C
	Central riser, ceiling branches
Balancing Strategy	Load split by window area & room zoning

The total calculated heating load of 20.38 kW split into 13.42 kW for the ground floor Figure 22 and 6.96 kW for the upper floor Figure 23, was increased by a 20% safety factor to ensure resilience against peak conditions. Redistribution of 10% of the upper-floor load to the ground-floor circuit successfully mitigated stratification and optimized thermal balance. Measured pressure drops of 4 317 Pa in the supply loop Table 18 and 4 789 Pa in the return loop Table 19 yielded a combined system resistance of 9 106 Pa, establishing a required pump dynamic head of 0.93 m at a nominal flow of 1.151 L/s (achieving 1.2 m/s velocity). The selected friction limit of 150 Pa/m maintained acceptable gradients, while copper piping with a thermal expansion coefficient of 17×10^{-6} /°C guaranteed dimensional stability and long-term reliability under cyclic temperature variations.

5. Conclusions and Future Work

This renovation project successfully demonstrates how integrating ventilation, daylighting, and energy strategies enhances overall building efficiency through an integrated design process. Using different approaches such as retrofitting windows, implementing active and passive shading techniques, and improving the thermal insulation properties of specific components of the building, an old building can become a new, improved version of itself, completing with necessary standards to make the building legal, as well as pleasant to work in every week day.

In terms of future work, further emphasis can be placed on the reuse of ventilation ducts and ventilation fittings. Moreover, incorporating renewable energy sources such as PV panel integration is recommended to maximise the building's potential and user satisfaction, as well as the owners' satisfaction.

Author Contributions:

Declaration Of Generative AI And AI-Assisted Technologies in The Writing Process

During the preparation of this work, the authors utilised ChatGPT and Deepseek.com to enhance readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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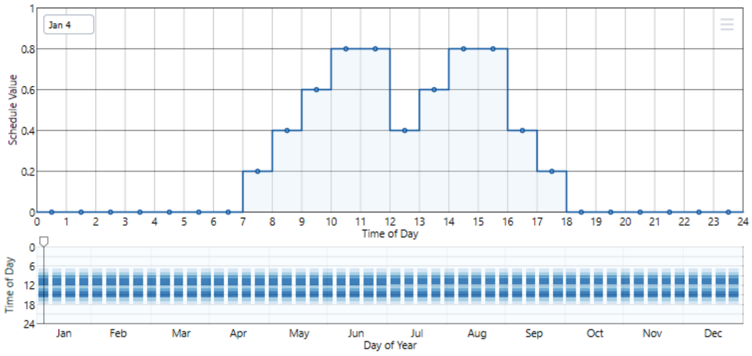
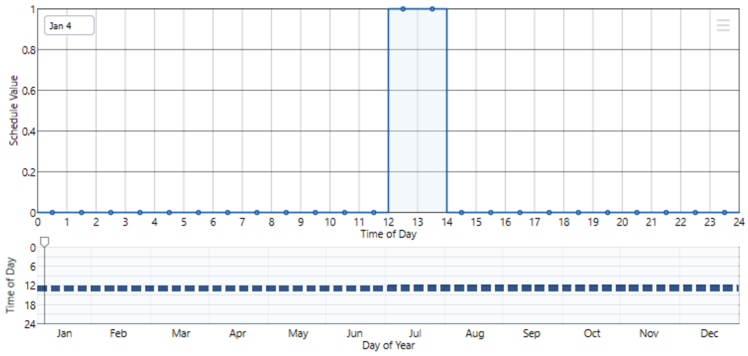
APPENDIX A- ZONE SCHEDULES

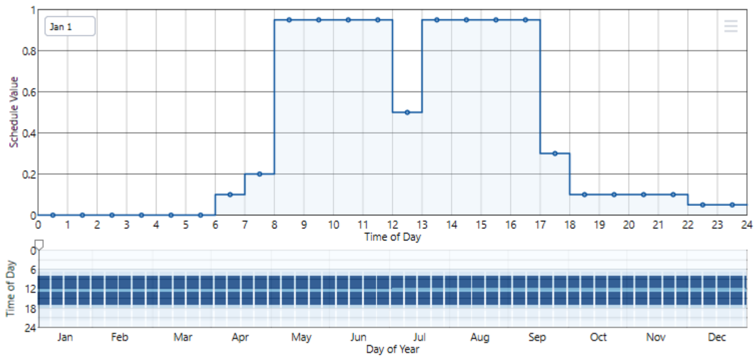
Zone 1- Equipment – Every workday



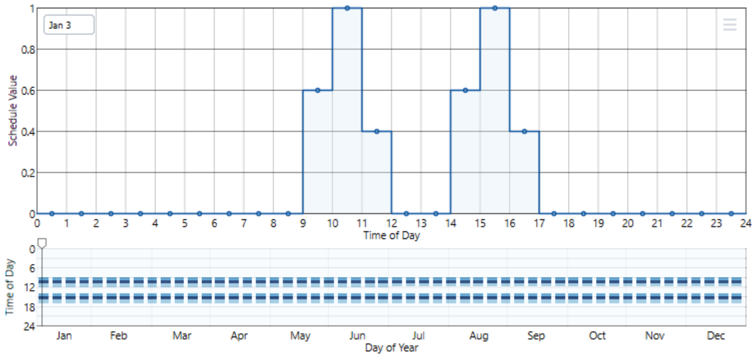
Zone 2 – People, Equipment, Lights – Every workday



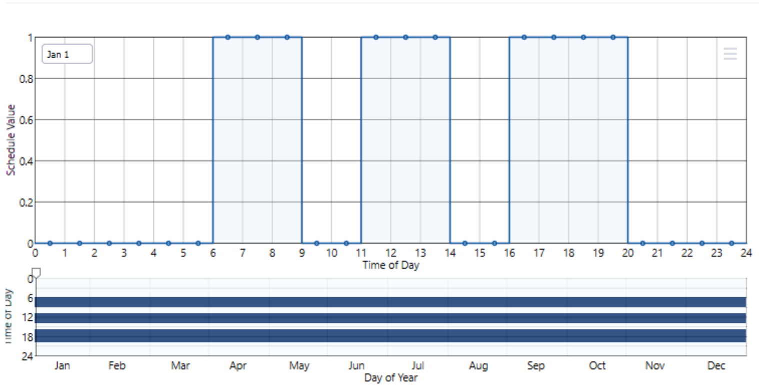
Zone 3 - People, Equipment, Lights – Every workday



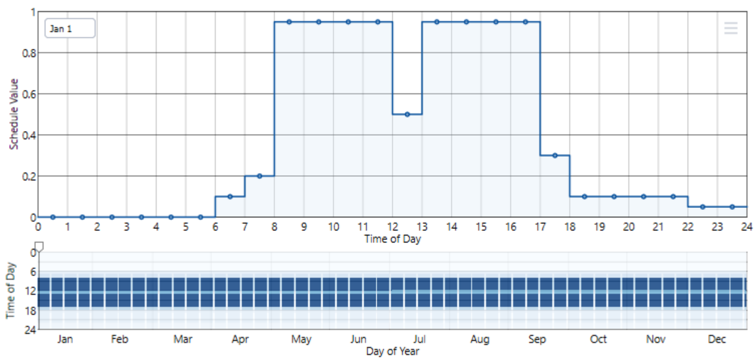
Zone 4 - People, Equipment, Lights – Every workday



Zone 5 – People, Lighting, Equipment is all on



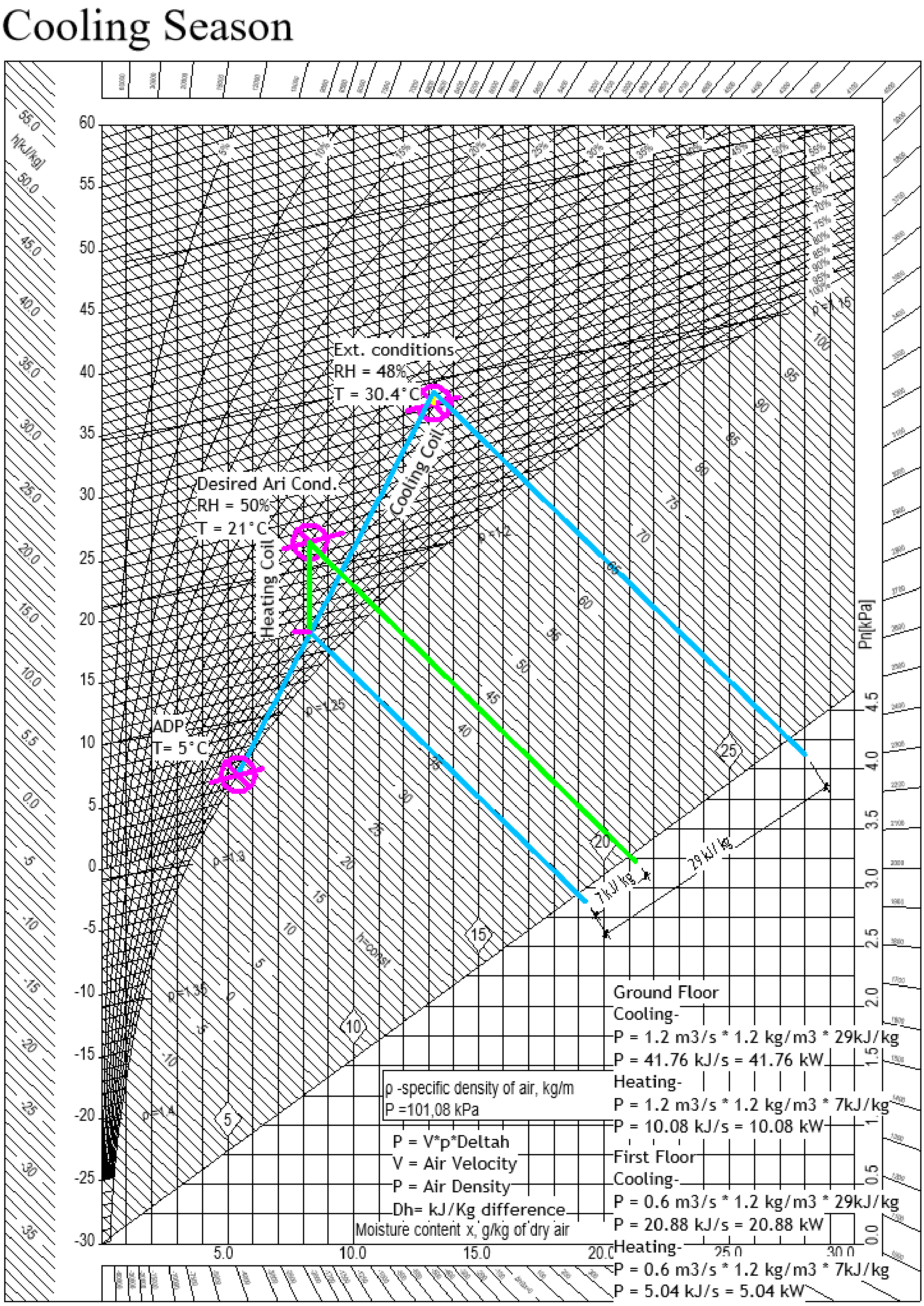
Zone 6 - People, Equipment, Lights – Every workday



Zone 7 - People, Equipment, Lights – Every workday

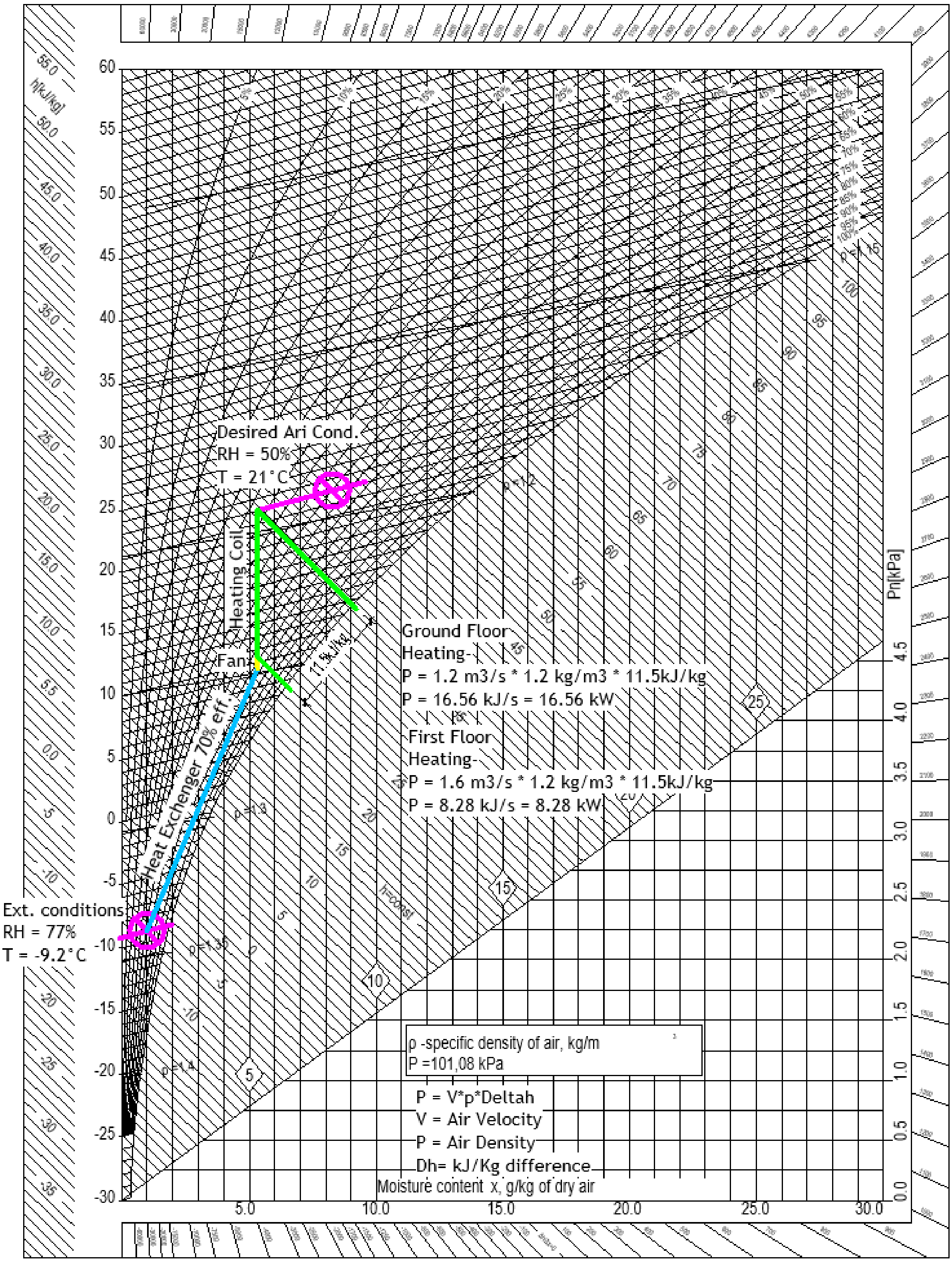
APPENDIX B- MOLLIER DIAGRAM

Cooling Season



Heating Season

Heating Season



APPENDIX C – DUCT LAYOUT

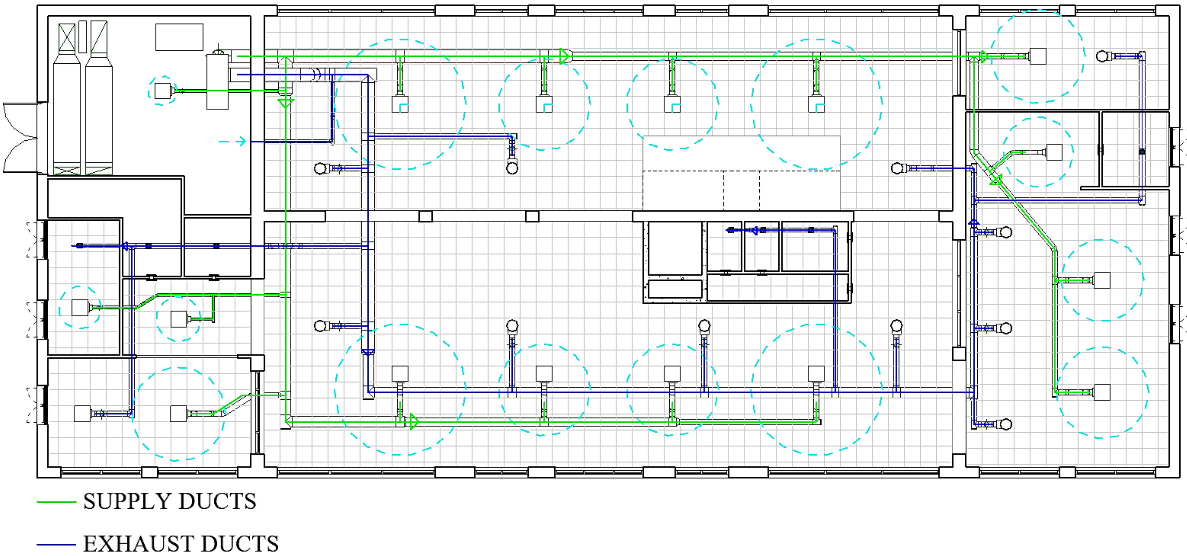


Figure 14. Ground Floor Ducting

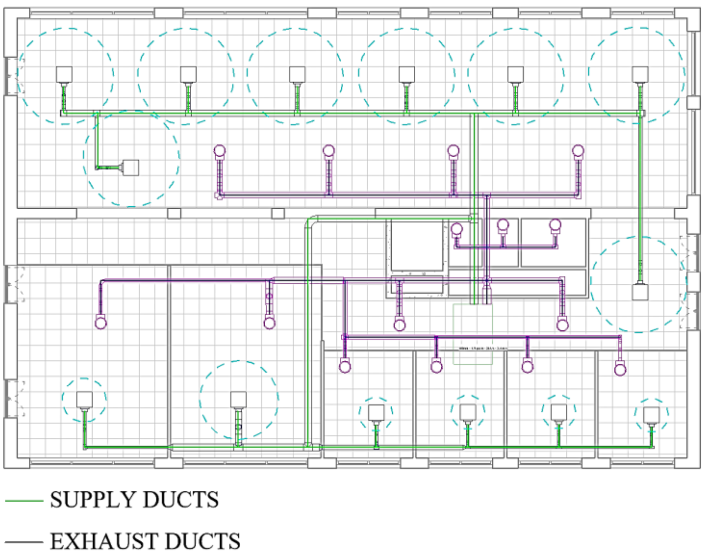


Figure 15. First Floor Ducting

APPENDIX D- SWEGON "RUD" AIRFLOW VISUALISATION

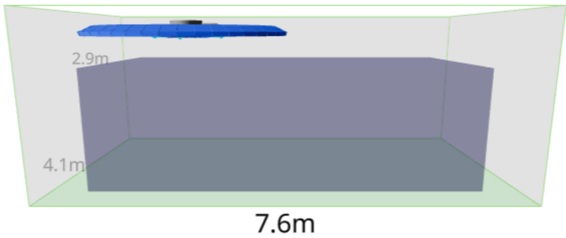


Figure 16 Ground-Floor- Meeting Room 1 And 2

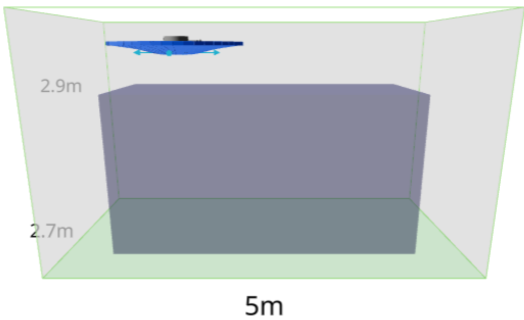


Figure 17. Ground-Floor- Rest Room

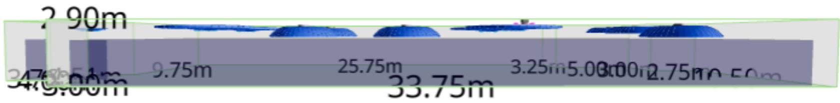


Figure 18. Ground-floor- Canteen and Open Office

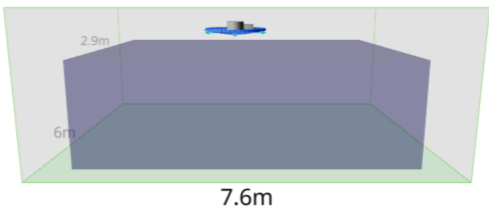


Figure 19. Ground-Floor- Technical Room

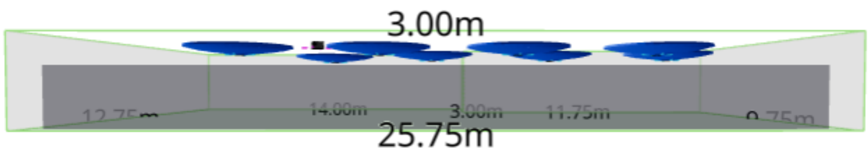


Figure 20. First Floor

APPENDIX E – RADIATOR HEATING SYSTEM

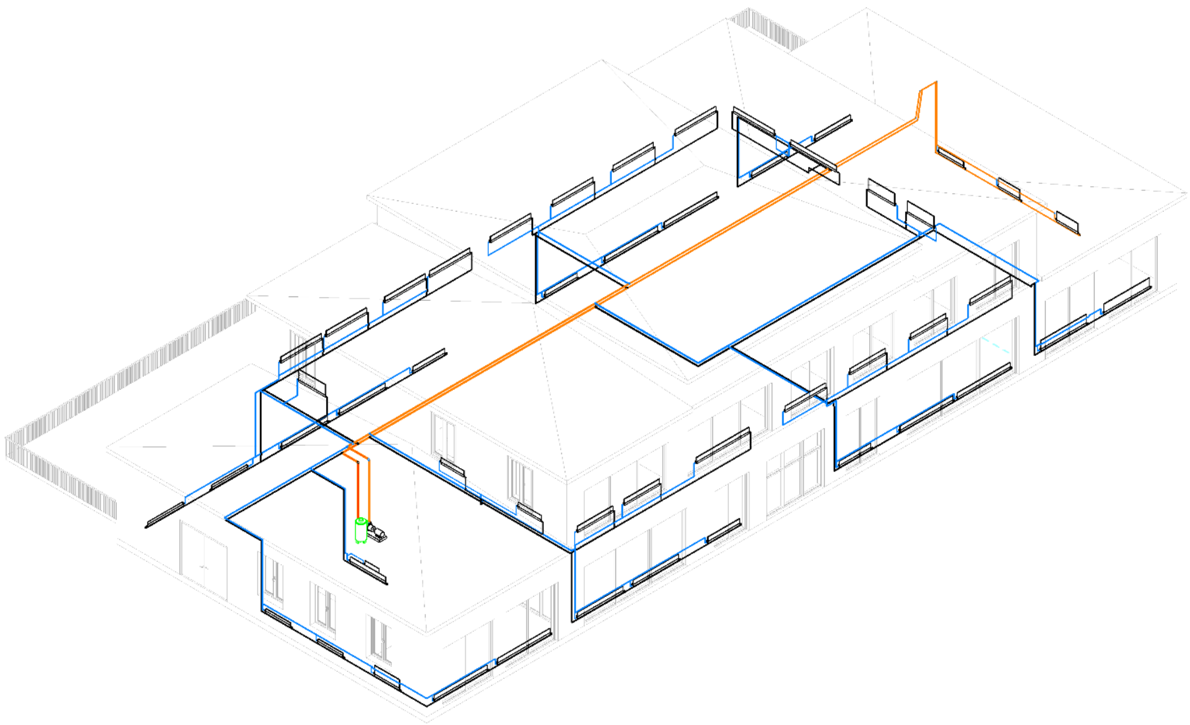


Figure 21 Radiator grid with critical path.

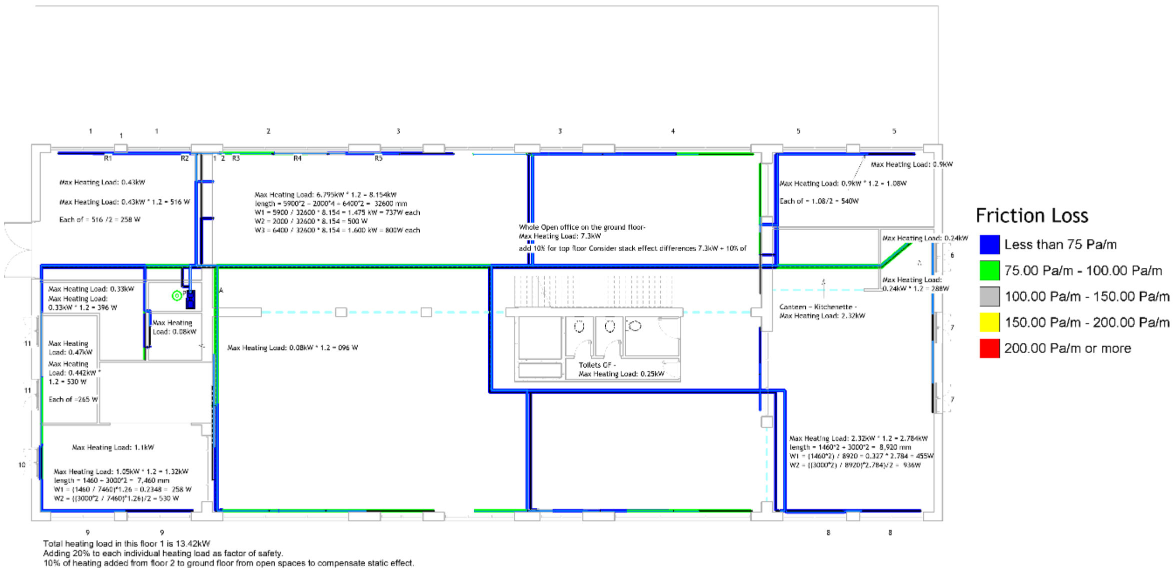


Figure 22 Ground floor plan with friction loss grading

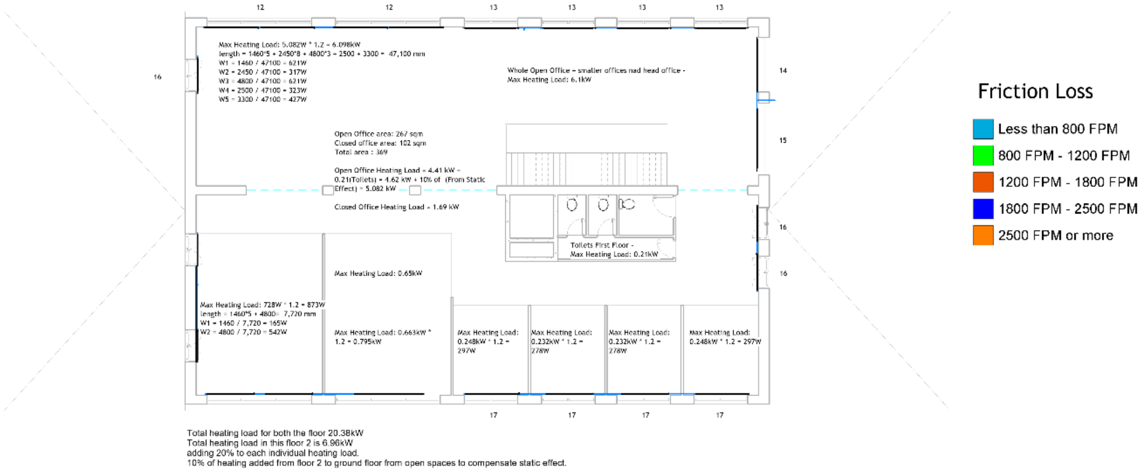


Figure 23 First Floor plan with Friction Loss grading.

Table 17 Radiator Selection and Sizing.

RADIATOR SYSTEM								
WINDOW	NO	NOS	WATT	WATT ACHIVED	TOTAL HEATING	LENGTH	WIDTH	HIGHT
3000	1	2	516	259	518.0	2000	22	200
5900	2	4	737	731	2924.0	2600	21	300
2000	3	4	500	506	2024.0	1800	33	300
6400	4	4	800	844	3376.0	3000	33	300
3000	5	2	540	562	1124.0	2000	33	300
1460	6	1	288	394	394.0	1400	33	300
1460	7	2	455	457	914.0	1200	33	500
3000	8	2	936	991	1982.0	2600	33	500
3000	9	2	530	536	1072.0	3000	22	300
1460	10	1	256	250	250.0	1400	22	300
1460	11	2	265	278	556.0	1400	33	200
Storage Room	1	1	396	398	398.0	2000	33	200
Server Room	1	1	96	104	104.0	800	33	200
4800	12	2	621	647	1294.0	2300	33	300
2450	13	4	317	335	1340.0	2300	21	300
2500	14	1	323	335	335.0	2300	21	300
3300	15	1	427	464	464.0	2600	22	300
1460	16	3	621	655	1965.0	1500	33	600
4800	17	1	795	794	794.0	2600	33	400
1460	18	2	165	190	380.0	1300	21	300
4800	19	2	542	562	1124.0	2000	33	300
4800	20	1	795	844	844.0	3000	33	300
2450	21	2	297	335	670.0	2300	21	300
2450	22	2	278	292	584.0	2000	21	300
					25430			

Table 18 Hydronic Supply Pressure Loss

Section	Element	Flow	Size	Velocity	Velocity Pressure	Length	K Coefficient	Friction	Total Pressure Loss	Section Pressure Loss
21.0	Pipe	1.049 L/s	54 mmø	0.5 m/s	-	200.0	-	63.25 Pa/m	12.69 Pa	154.22 Pa
	Fittings	1.049 L/s	-	0.5 m/s	125.81 Pa	-	1.1	-	141.53 Pa	~
22.0	Fittings	0.766 L/s	-	0.4 m/s	67.09 Pa	-	0.6	-	66.05 Pa	66.05 Pa
23.0	Pipe	0.766 L/s	42 mmø	0.6 m/s	-	893.0	-	128.26 Pa/m	115.14 Pa	115.14 Pa
	Fittings	0.766 L/s	-	0.6 m/s	193.41 Pa	-	0.0	-	0.00 Pa	~
24.0	Pipe	0.581 L/s	42 mmø	0.5 m/s	-	12809.0	-	78.96 Pa/m	1015.59 Pa	1060.88 Pa
	Fittings	0.581 L/s	-	0.5 m/s	111.07 Pa	-	0.4	-	45.29 Pa	~
25.0	Fittings	0.291 L/s	-	0.2 m/s	27.87 Pa	-	0.6	-	21.07 Pa	21.07 Pa
26.0	Pipe	0.291 L/s	35 mmø	0.3 m/s	-	1852.0	-	59.67 Pa/m	110.92 Pa	110.92 Pa
	Fittings	0.291 L/s	-	0.3 m/s	60.68 Pa	-	0.0	-	0.00 Pa	~
27.0	Fittings	0.139 L/s	-	0.2 m/s	13.84 Pa	-	0.6	-	12.79 Pa	12.79 Pa
28.0	Pipe	0.139 L/s	28 mmø	0.3 m/s	-	11430.0	-	48.63 Pa/m	557.57 Pa	557.57 Pa
	Fittings	0.139 L/s	-	0.3 m/s	34.21 Pa	-	0.0	-	0.00 Pa	~
29.0	Fittings	0.056 L/s	-	0.1 m/s	5.61 Pa	-	0.7	-	13.91 Pa	13.91 Pa
30.0	Pipe	0.056 L/s	18 mmø	0.3 m/s	-	10762.0	-	77.92 Pa/m	838.55 Pa	870.07 Pa
	Fittings	0.056 L/s	-	0.3 m/s	39.10 Pa	-	0.8	-	31.51 Pa	~
334.0	Pipe	0.020 L/s	12 mmø	0.3 m/s	-	3558.0	-	104.53 Pa/m	371.93 Pa	376.05 Pa
	Fittings	0.020 L/s	-	0.3 m/s	31.28 Pa	-	0.0	-	4.12 Pa	~
	Equipment	0.020 L/s	-	-	-	-	-	-	0.00 Pa	~
335.0	Fittings	0.020 L/s	-	0.1 m/s	10.95 Pa	-	0.7	-	12.01 Pa	12.01 Pa
336.0	Pipe	0.039 L/s	15 mmø	0.3 m/s	-	3360.0	-	101.65 Pa/m	341.52 Pa	341.52 Pa
	Fittings	0.039 L/s	-	0.3 m/s	43.80 Pa	-	0.0	-	0.00 Pa	~
337.0	Fittings	0.039 L/s	-	0.2 m/s	19.09 Pa	-	1.7	-	36.02 Pa	36.02 Pa

Critical Path : 21-22-23-24-25-26-27-28-29-30-334-335-336-337-342 ; Total Pressure Loss : 4317.31 Pa

Table 19 Hydronic Return Pressure Loss

Section	Element	Flow	Size	Velocity	Velocity Pressure	Length	K Coefficient	Friction	Total Pressure Loss	Section Pressure Loss
1.0	Fittings	1.151 L/s	-	3.7 m/s	6712.83 Pa	-	0.2	-	35.15 Pa	35.15 Pa
2.0	Pipe	1.151 L/s	54 mmø	0.6 m/s	-	4447.0	-	74.46 Pa/m	332.57 Pa	588.22 Pa
	Fittings	1.151 L/s	-	0.6 m/s	151.50 Pa	-	1.7	-	255.64 Pa	~
35.0	Pipe	0.056 L/s	18 mmø	0.3 m/s	-	12879.0	-	77.92 Pa/m	1003.57 Pa	1064.79 Pa
	Fittings	0.056 L/s	-	0.3 m/s	39.10 Pa	-	1.5	-	61.22 Pa	~
36.0	Fittings	0.056 L/s	-	0.1 m/s	5.61 Pa	-	1.6	-	46.06 Pa	46.06 Pa
37.0	Pipe	0.139 L/s	28 mmø	0.3 m/s	-	11539.0	-	48.78 Pa/m	562.89 Pa	562.89 Pa
	Fittings	0.139 L/s	-	0.3 m/s	34.21 Pa	-	0.0	-	0.00 Pa	~
38.0	Fittings	0.139 L/s	-	0.2 m/s	13.84 Pa	-	1.3	-	36.88 Pa	36.88 Pa
39.0	Pipe	0.291 L/s	35 mmø	0.3 m/s	-	1748.0	-	59.67 Pa/m	104.67 Pa	104.67 Pa
	Fittings	0.291 L/s	-	0.3 m/s	60.68 Pa	-	0.0	-	0.00 Pa	~
40.0	Fittings	0.291 L/s	-	0.2 m/s	27.87 Pa	-	1.2	-	58.38 Pa	58.38 Pa
41.0	Pipe	0.581 L/s	42 mmø	0.5 m/s	-	12809.0	-	78.96 Pa/m	1015.59 Pa	1060.88 Pa
	Fittings	0.581 L/s	-	0.5 m/s	111.07 Pa	-	0.4	-	45.29 Pa	~
42.0	Pipe	0.766 L/s	42 mmø	0.6 m/s	-	583.0	-	128.26 Pa/m	75.13 Pa	75.13 Pa
	Fittings	0.766 L/s	-	0.6 m/s	193.41 Pa	-	0.0	-	0.00 Pa	~
43.0	Fittings	0.766 L/s	-	0.4 m/s	67.09 Pa	-	1.4	-	215.31 Pa	215.31 Pa
44.0	Pipe	1.049 L/s	54 mmø	0.5 m/s	-	802.0	-	63.51 Pa/m	50.93 Pa	192.45 Pa
	Fittings	1.049 L/s	-	0.5 m/s	125.81 Pa	-	1.1	-	141.53 Pa	~
330.0	Fittings	0.039 L/s	-	0.2 m/s	19.09 Pa	-	1.2	-	40.86 Pa	40.86 Pa
331.0	Pipe	0.039 L/s	15 mmø	0.3 m/s	-	3160.0	-	101.65 Pa/m	321.19 Pa	321.19 Pa
	Fittings	0.039 L/s	-	0.3 m/s	43.80 Pa	-	0.0	-	0.00 Pa	~
332.0	Fittings	0.020 L/s	-	0.1 m/s	10.95 Pa	-	1.3	-	31.47 Pa	31.47 Pa
333.0	Pipe	0.020 L/s	12 mmø	0.3 m/s	-	3353.0	-	104.53 Pa/m	350.47 Pa	354.59 Pa
	Fittings	0.020 L/s	-	0.3 m/s	31.28 Pa	-	0.0	-	4.12 Pa	~
	Equipment	0.020 L/s	-	-	-	-	-	-	0.00 Pa	~

Critical Path : 1-2-35-36-37-38-39-40-41-42-43-44-330-331-332-333 ; Total Pressure Loss : 4788.91 Pa

APPENDIX F – ELECTRIC LIGHTING SYSTEM

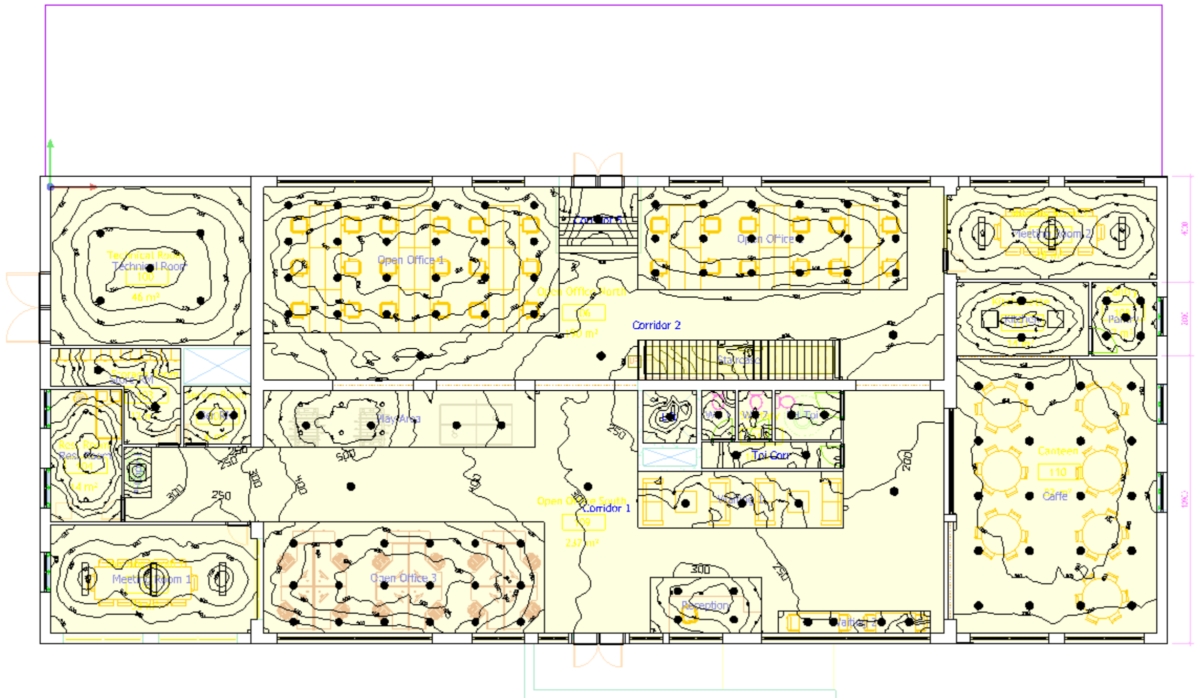


Figure 24. Ground Floor Lighting layout.

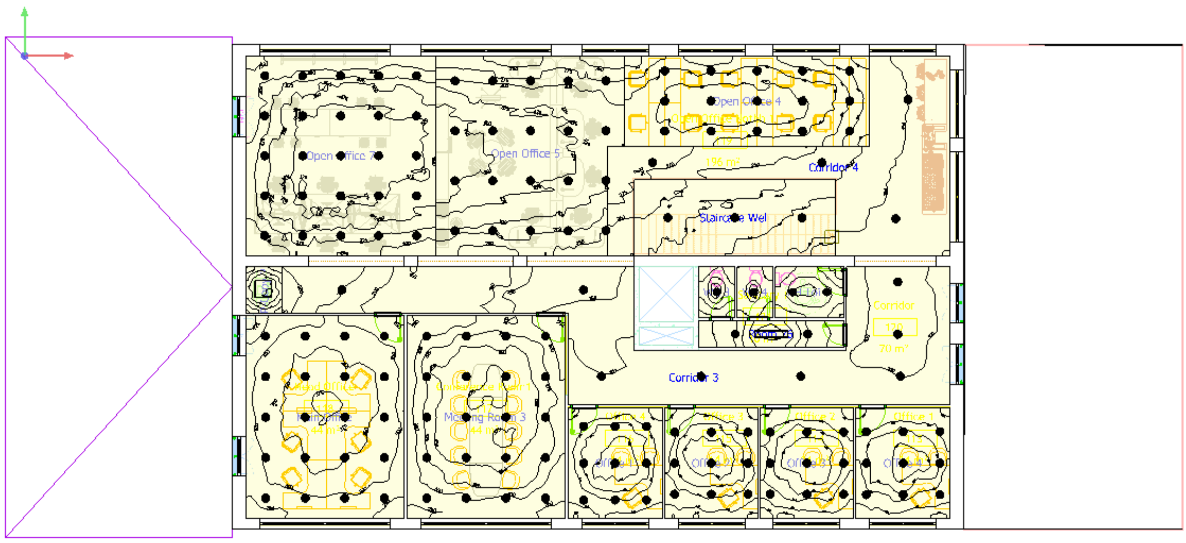


Figure 25. First Floor Lighting layout

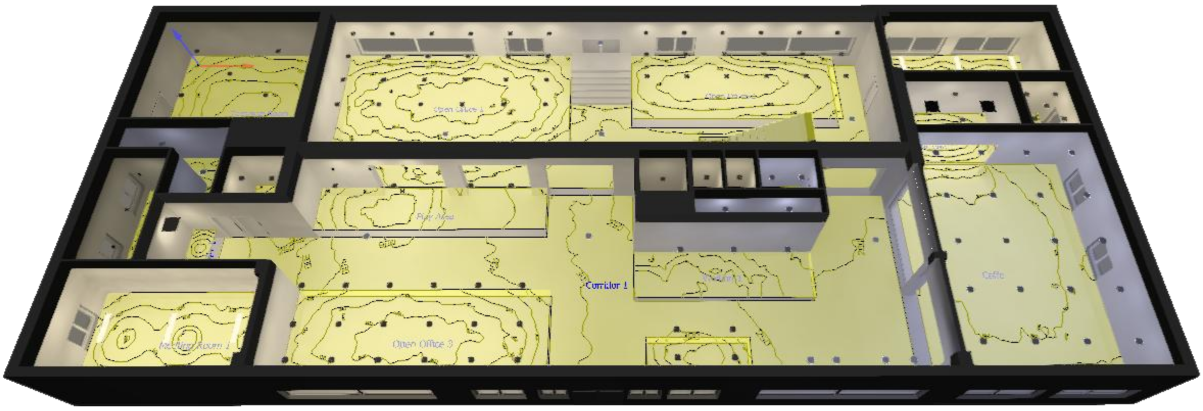


Figure 26. Ground Floor 3D Lighting Visual.

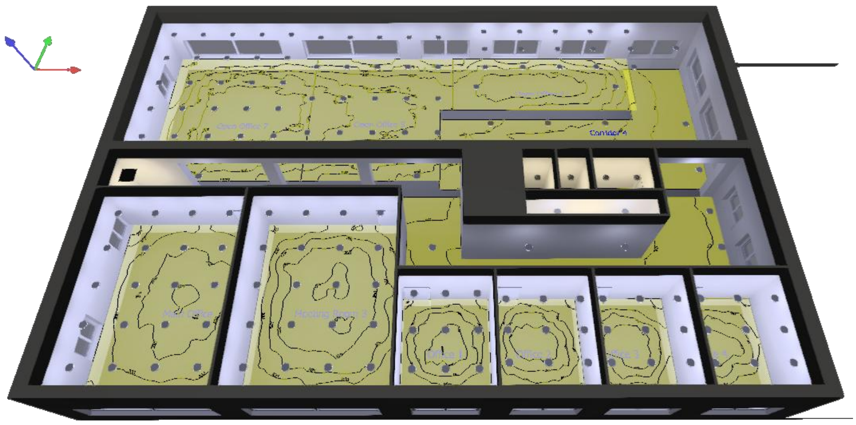


Figure 27. First Floor 3D Lighting Visual.

Table 20 First Floor Electric lighting Target Lux.

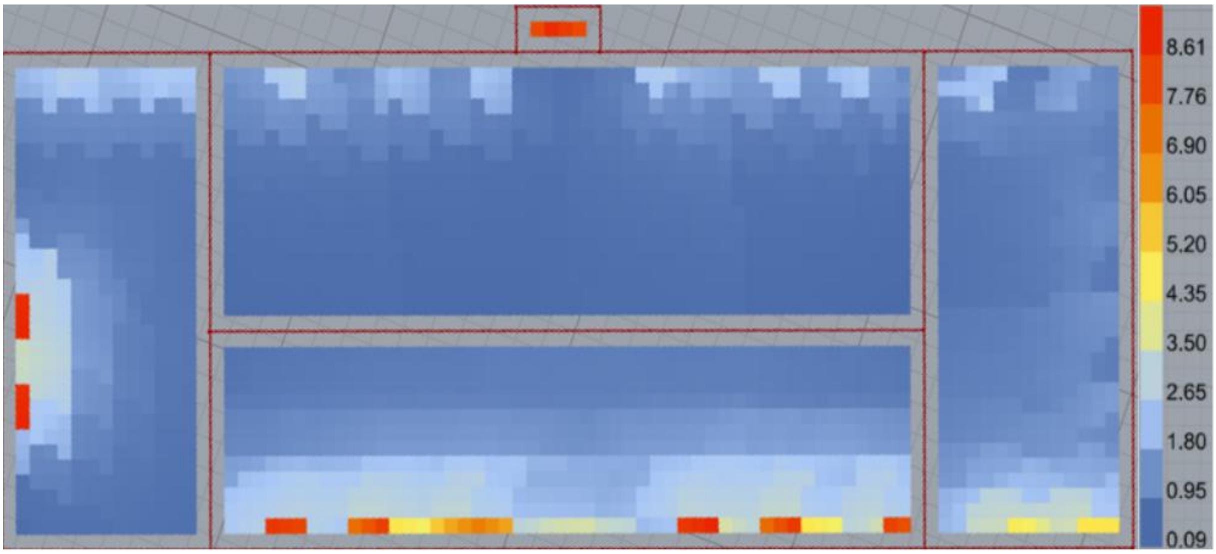
Room	Lux (lx)		Min / Average		LENI
	Actual	Target	Actual	Target	kWh/(m2*a)
Caffe	303.0	≥ 300	0.7	≥ 0.6	45879.0
Corridor 1	296.0	≥100	0.5	≥0.4	-
Corridor 2	373.0	≥100	1.0	≥0.5	1.0
Corridor 5	184.0	≥100	0.5	≥0.6	2.0
Kitchen	547.0	≥500	0.7	≥0.6	13.0
Lift	154.0	≥100	0.9	≥0.6	3.0
Meeting Room 1	563.0	≥500	0.6	≥0.6	45784.0
Meeting Room 2	565.0	≥500	0.6	≥0.6	5 – 8
Open Office 1	583.0	≥500	0.7	≥0.6	12.0
Open Office 2	504.0	≥500	0.9	≥0.6	13.0
Open Office 3	537.0	≥500	0.7	≥0.6	12.0
Pantry	352.0	≥300	0.6	≥0.6	45882.0
P-Copy 1	456.0	≥300	0.9	≥0.4	1.0
PH – Toilet 1	214.0	≥200	0.6	≥0.4	4.0
Play Area	459.0	≥300	0.6	≥0.4	7.0
Reception	347.0	≥300	0.8	≥0.6	12.0
Rest Room	126.0	≥100	0.8	≥0.4	45721.0
Server Room	315.0	≥300	0.8	≥0.6	7.0
Staircase	248.0	≥100	0.5	≥0.4	-
Store Room	108.0	≥100	0.8	≥0.4	4.0
Technical Room	204.0	≥200	0.7	≥0.4	0.0
Toilet Corridor 1	119.0	≥100	0.8	≥0.4	4.0
Waiting 1	238.0	≥200	0.8	≥0.4	4.0
Waiting 2	235.0	≥200	0.9	≥0.4	13.0
WC 1	234.0	≥200	0.9	≥0.4	4.0
WC 2	236.0	≥200	0.9	≥0.4	4.0

Table 21 Second Floor Electric lighting Target Lux.

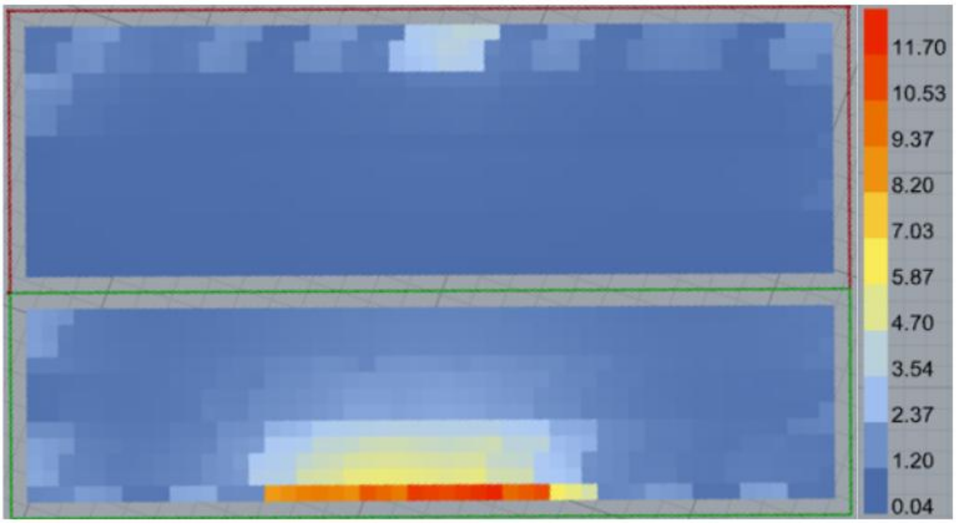
Room	Lux (lx)		Min/average		LENI
	Actual	Target	Actual	Target	kWh/(m2*a)
Corridor 3	125.0	>100	0.5	>0.4	1.0
Corridor 4	182.0	>100	0.5	>0.4	1.0
Main Office	319.0	>300	0.7	>0.6	45754.0
Meeting Room 3	326.0	>300	0.7	>0.6	45754.0
Office 1	355.0	>300	0.8	>0.6	45786.0
Office 2	354.0	>300	0.8	>0.6	45786.0
Office 3	356.0	>300	0.8	>0.6	45786.0
Office 4	352.0	>300	0.8	>0.6	45786.0
Open Office 4	301.0	>300	0.7	>0.6	8.0
Open Office 5	308.0	>300	0.7	>0.6	7.0
Open Office 6	349.0	>300	0.7	>0.6	8.0
P-Copy 2	487.0	>300	0.9	>0.4	2.0
PH – Toilet 2	287.0	>200	0.8	>0.4	4.0
Toilet Corridor 1	406.0	>100	0.6	>0.4	19.0
WC 3	234.0	>200	0.9	>0.4	4.0
WC 4	236.0	>200	0.9	>0.4	4.0

APPENDIX G – DAYLIGHTING- BASE CASE

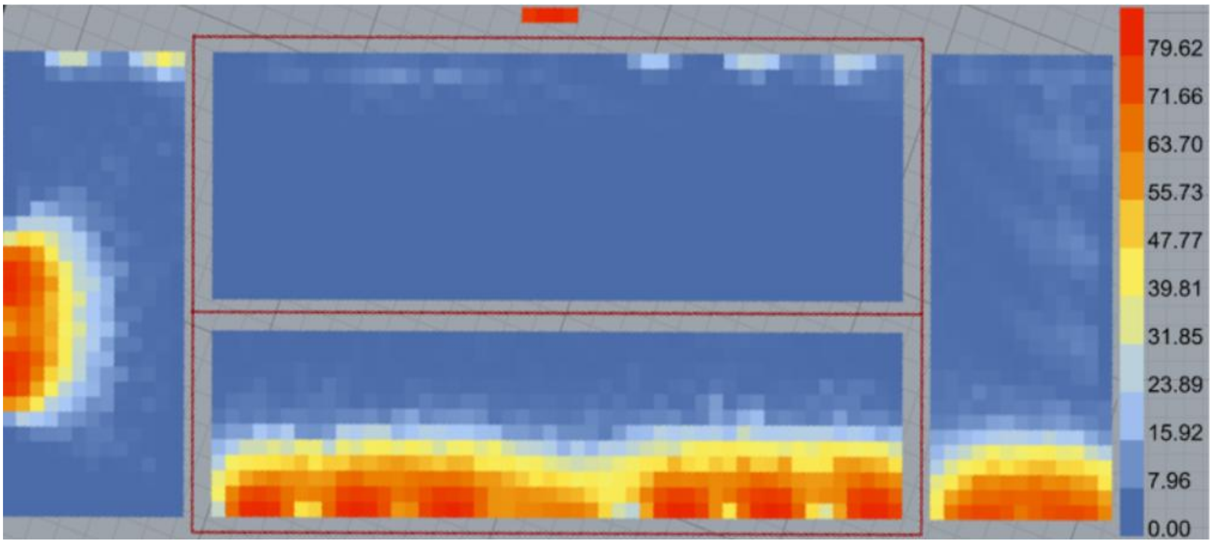
(DF) Daylight Factor - GROUNG FLOOR



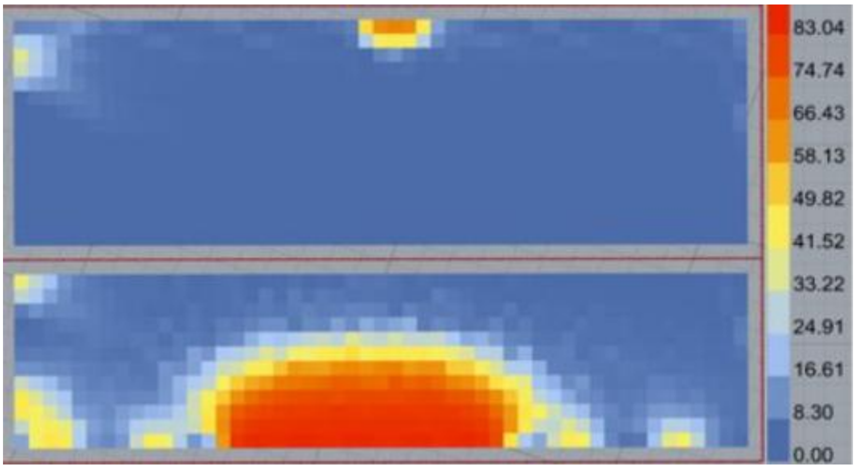
(DF) Daylight Factor - FIRST FLOOR



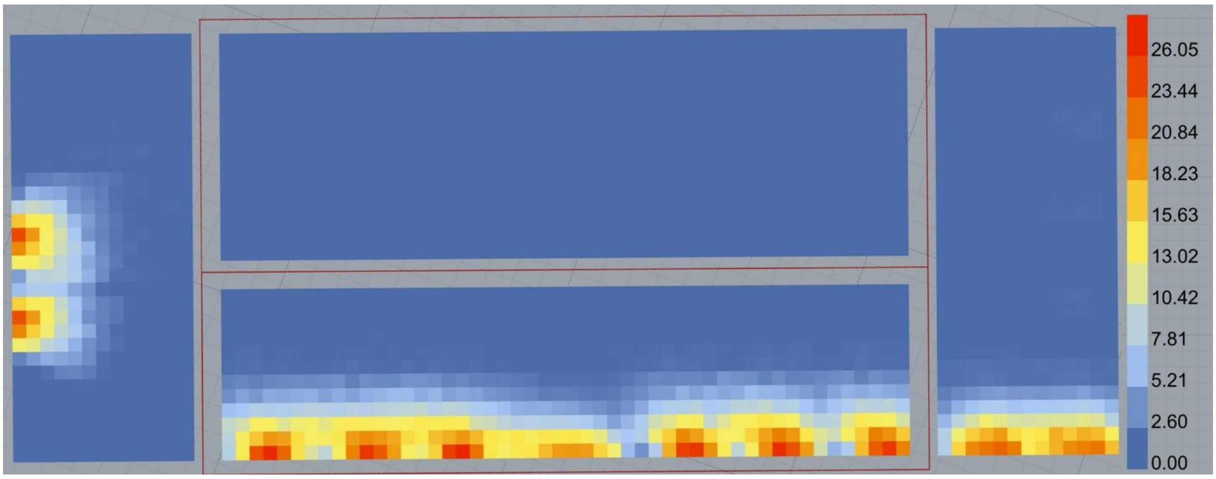
(sDA) Spatial Daylight Autonomy - GROUND FLOOR



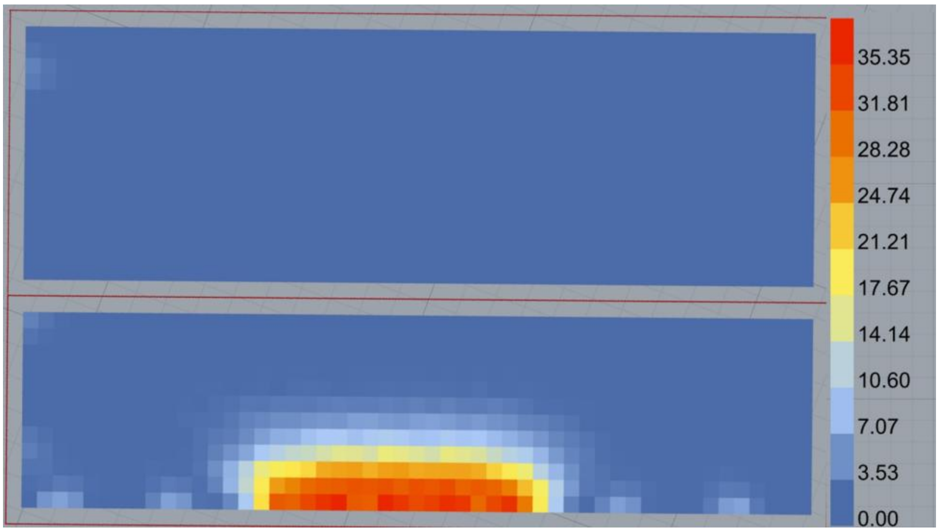
(sDA) Spatial Daylight Autonomy - FIRST FLOOR



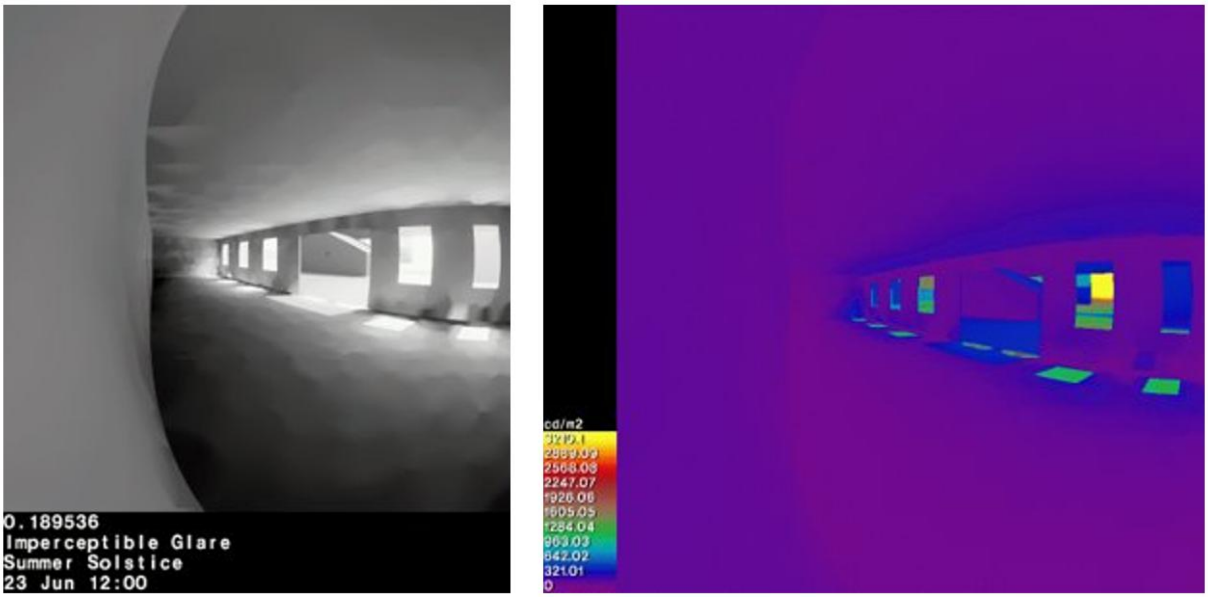
(DGP) Annual Daylight Glare Probability - Heat-Map Based- GROUND FLOOR



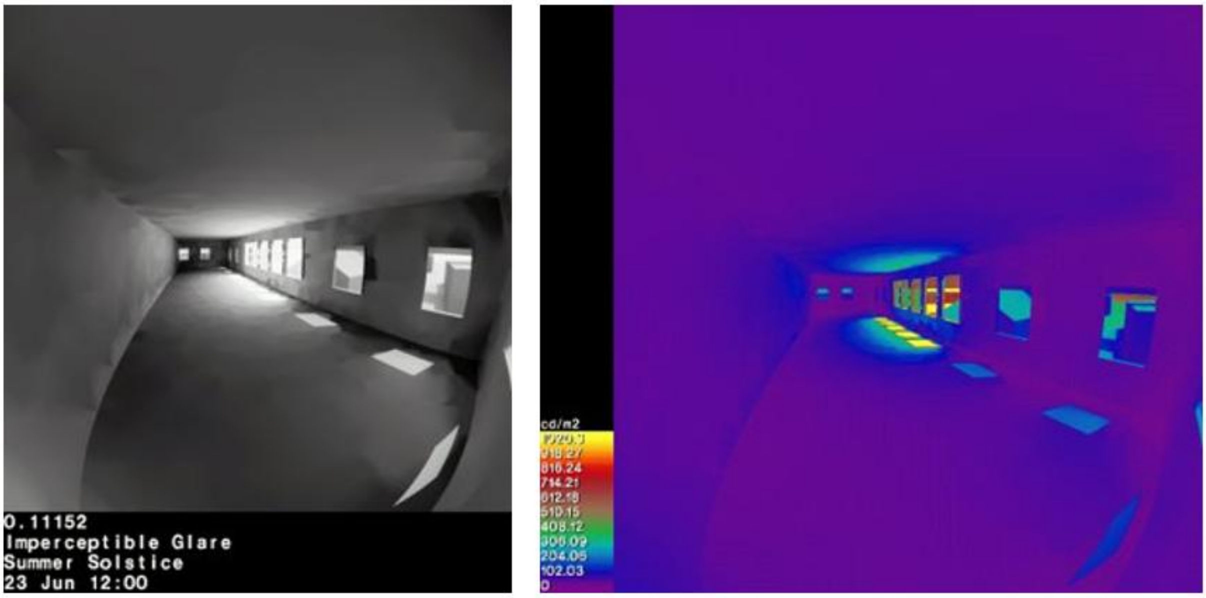
(DGP) Annual Daylight Glare Probability - Heat-Map Based- FIRST FLOOR



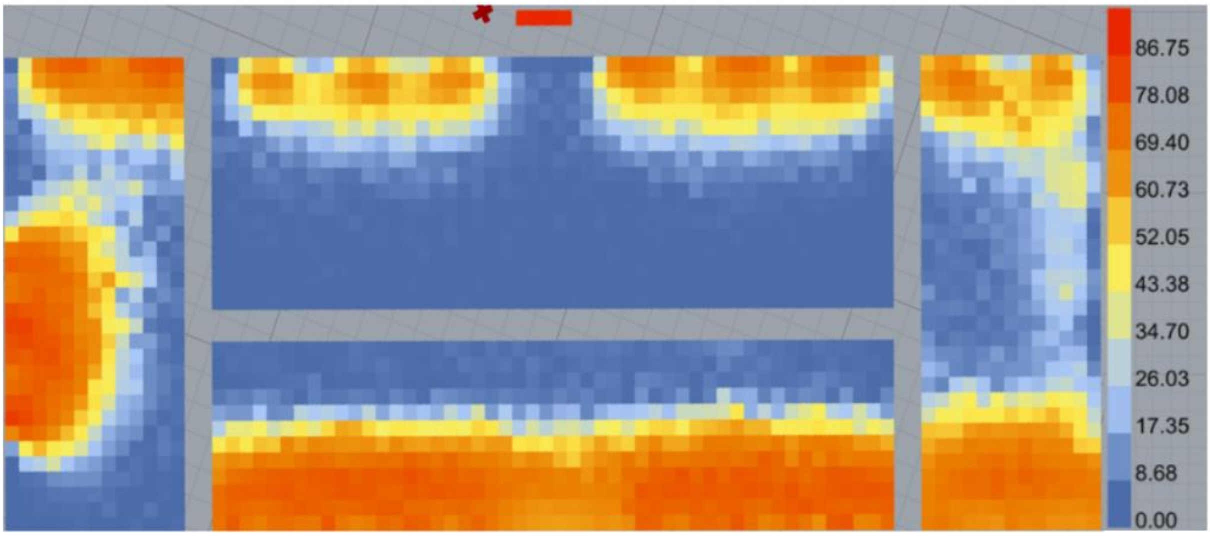
(DGP) Daylight Glare Probability – Visual Based - GROUND FLOOR



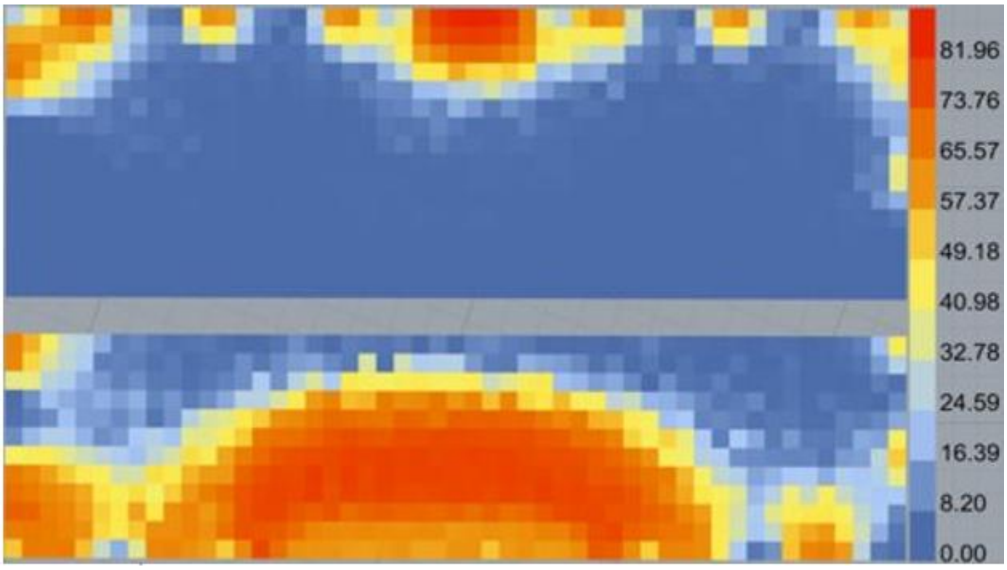
(DGP) Daylight Glare Probability – Visual Based - FIRST FLOOR



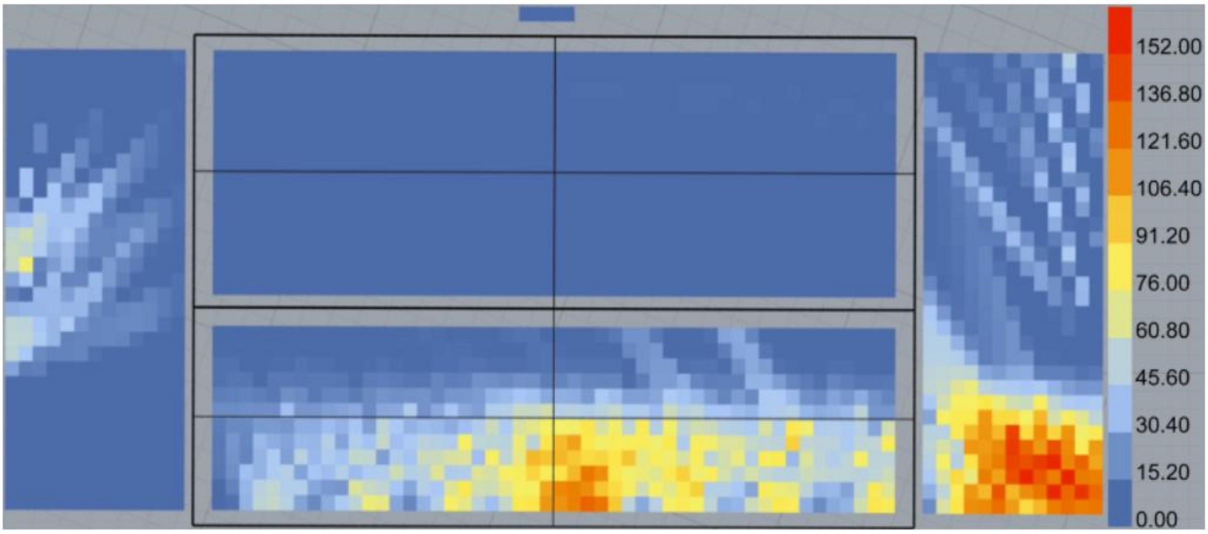
(UDI) Useful Daylight Illuminance - GROUND FLOOR



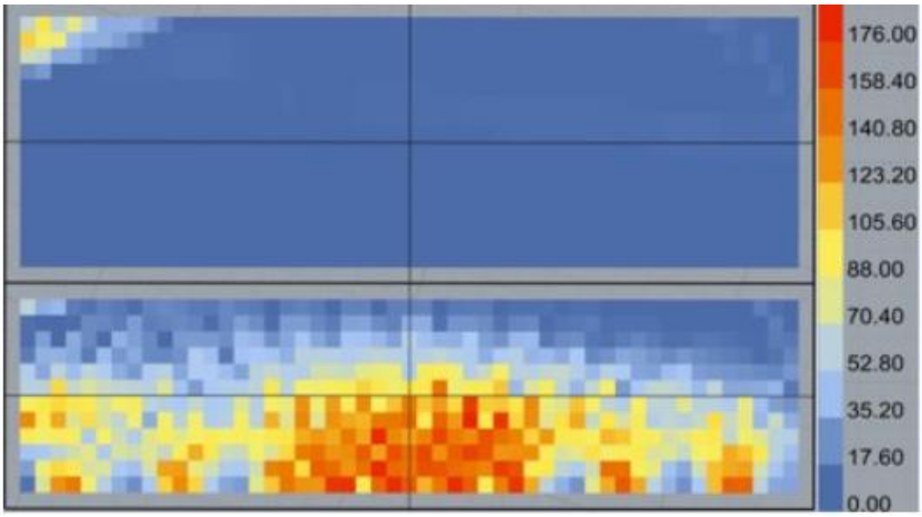
(UDI) Useful Daylight Illuminance - FIRST FLOOR



Direct Sun Exposure - GROUND FLOOR

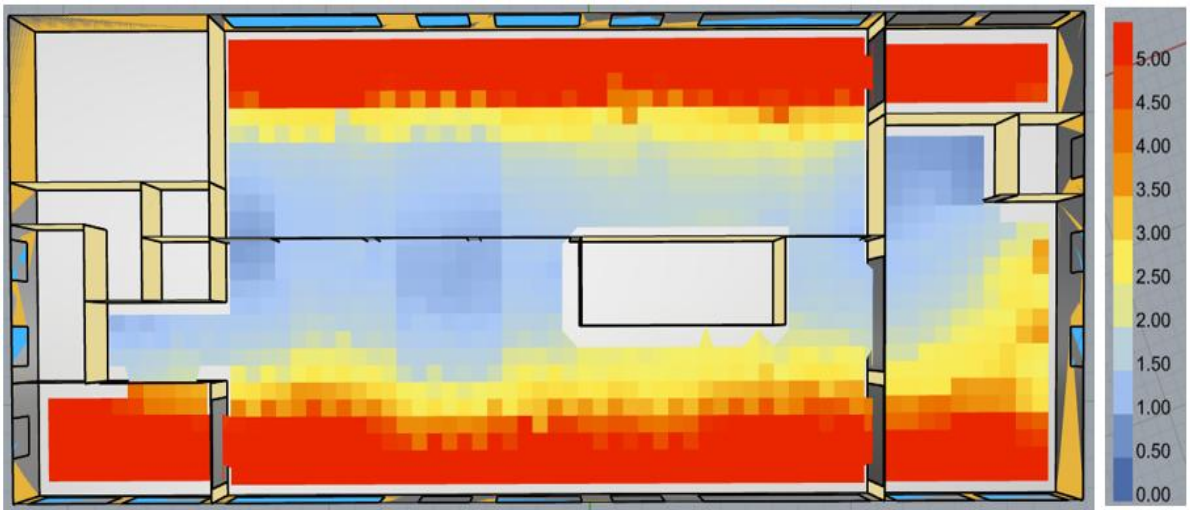


Direct Sun Exposure - FIRST FLOOR

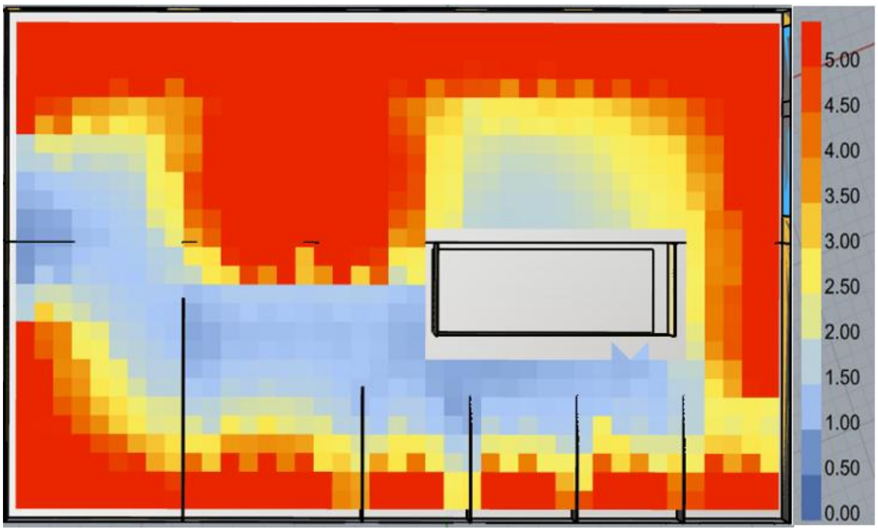


APPENDIX H – DAYLIGHTING- IMPROVED CASE

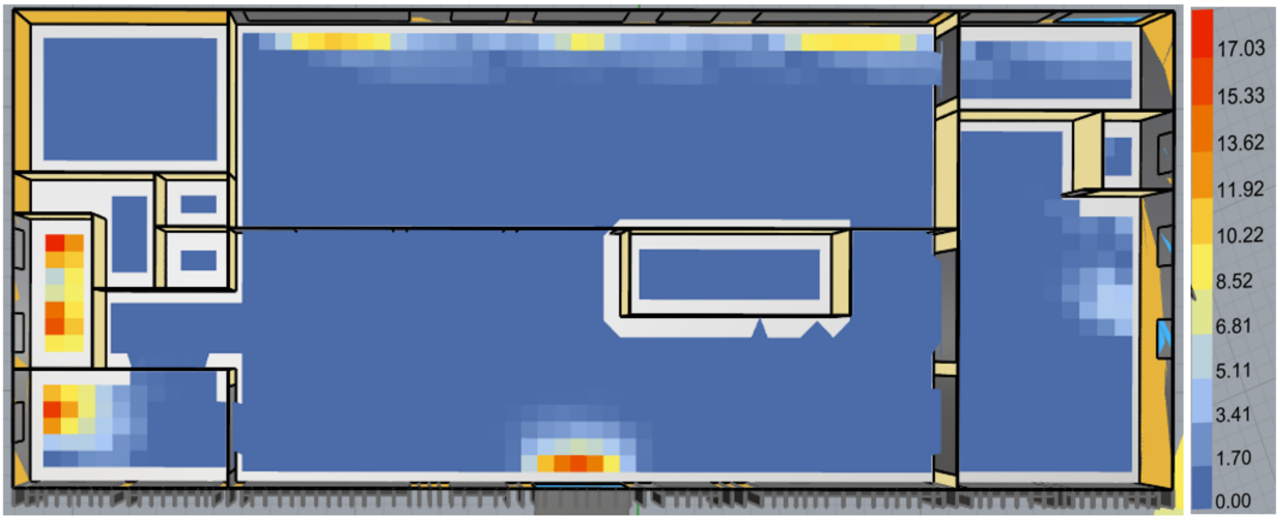
(DF) Daylight Factor- GROUND FLOOR



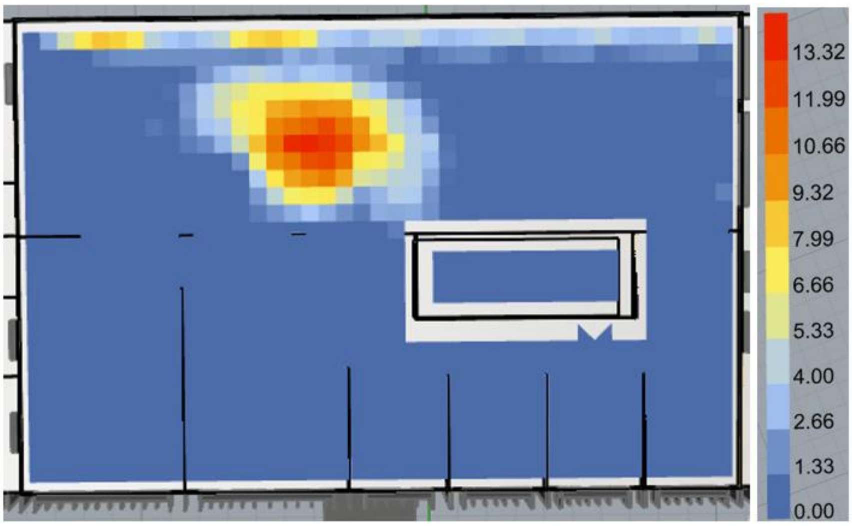
(DF) Daylight Factor- FIRST FLOOR



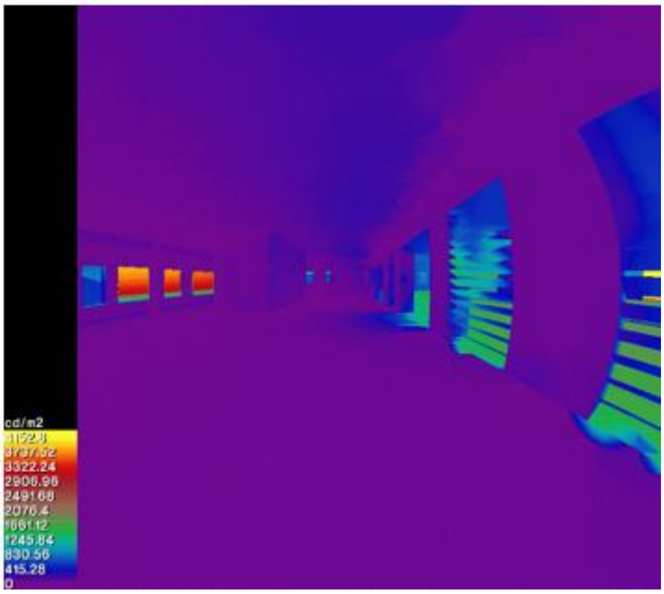
(DGP) Annual Daylight Glare Probability - Heat-Map Based- GROUND FLOOR



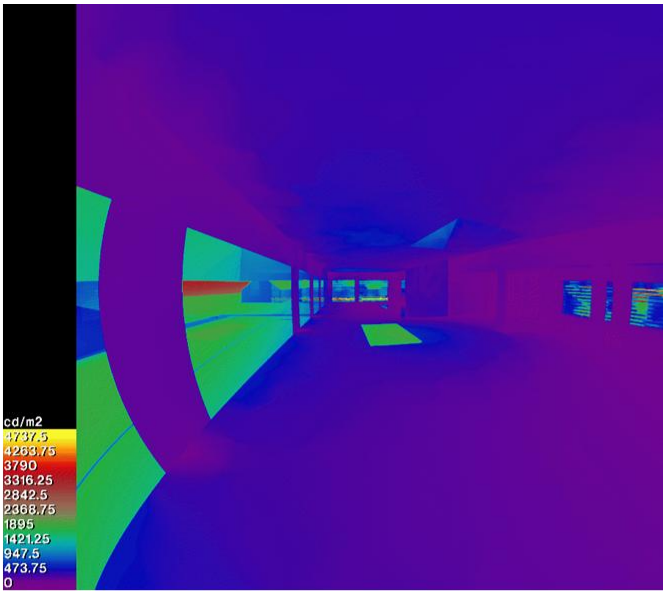
(DGP) Annual Daylight Glare Probability - Heat-Map Based- FIRST FLOOR



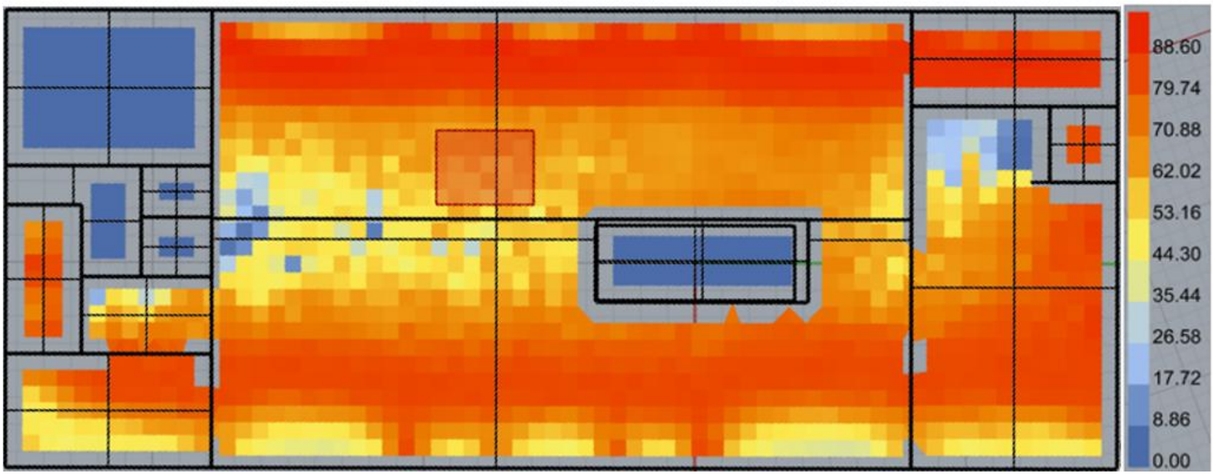
(DGP) Daylight Glare Probability – Visual Based - GROUND FLOOR



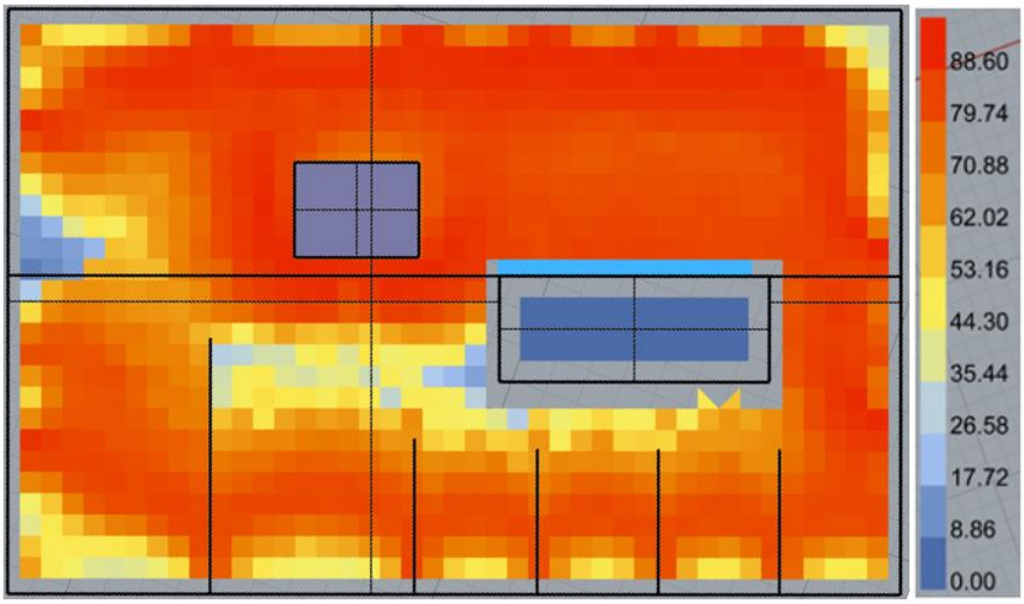
(DGP) Daylight Glare Probability – Visual Based - FIRST FLOOR



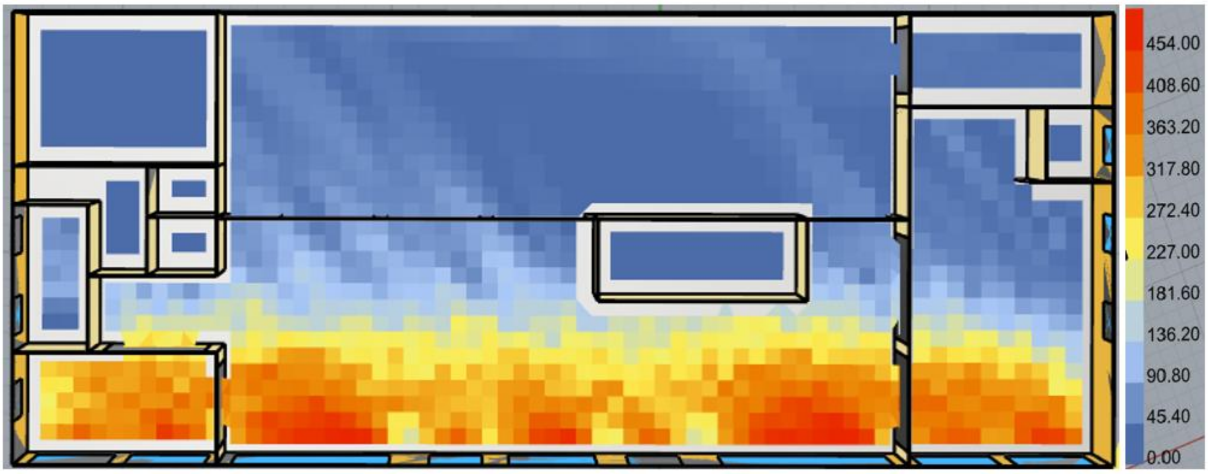
(UDI) Useful Daylight Illuminance - GROUND FLOOR



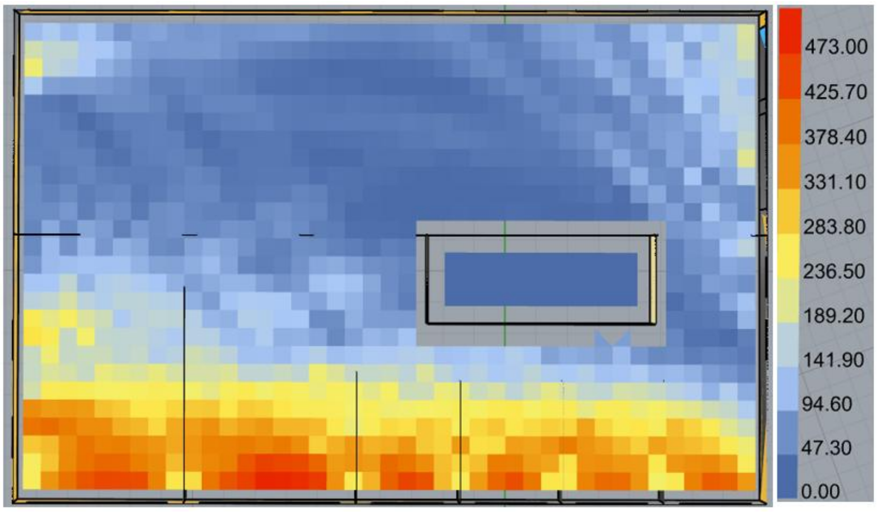
(UDI) Useful Daylight Illuminance - FIRST FLOOR



Direct Sun Exposure - GROUND FLOOR- Average of 116.9 hours



Direct Sun Exposure - FIRST FLOOR – Average of 113.14 hours



APPENDIX I – FLOOR PLANS, ELEVATIONS, AND SECTIONS

Figure 28. Proposed Ground Floor Plan.

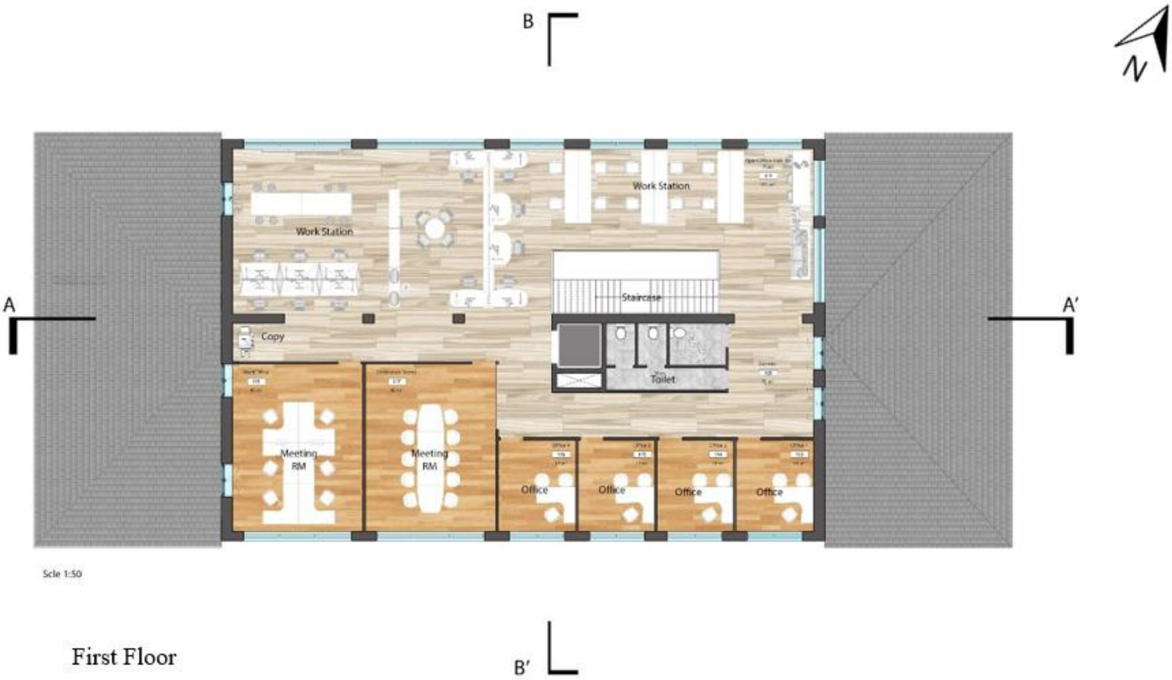


Figure 29. Proposed First Floor Plan.

Figure 30. Section at B-B'

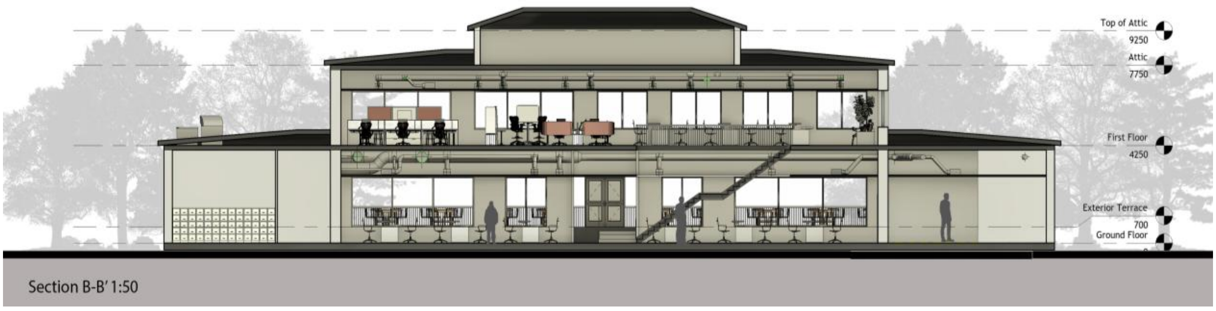


Figure 31. Section at A-A'

Figure 32. East Elevation

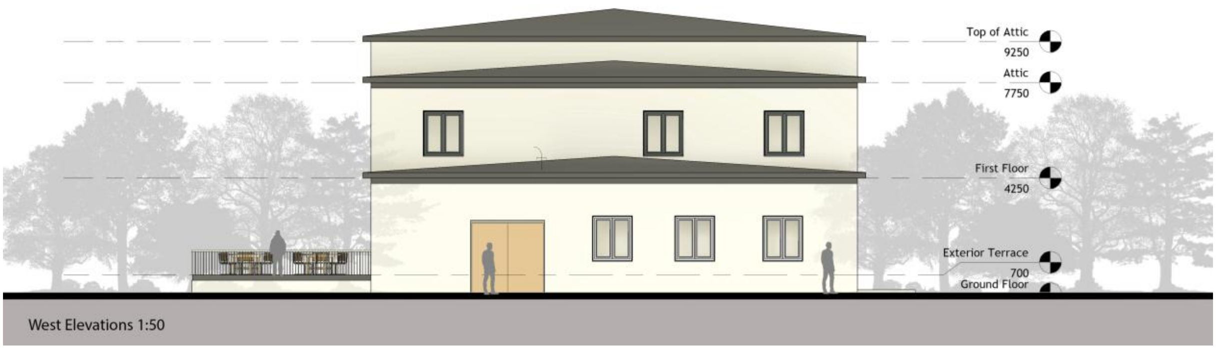


Figure 33. West Elevation

Figure 34. South Elevation

Figure 35. North Elevation